portfolio

Martin He

[selected Programming, HCI, and Architecture works]

Carnegie Mellon University SOA B.Arch

Martin He

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Education

Carnegie Mellon University
Pittsburgh, PA
Bachelor of Architecture | Dec 2020 [with College Honors]

Carnegie Mellon University
Pittsburgh, PA
Non-Degree Student | Jan 2024-Present

Relevant Classwork

Relevant CS Coursework: Imperative Computation, Fundamental of Programming, Concepts of Math, Computer System (expected), Math for ML (expected), Introduction to ML (expected)

Relevant Architecture Coursework: Architecture Studio, Parametric Design, Accounting, Real Estate Analysis, Micro-Economics

Experience

EOA Architects, PLLC
Architect | Nashville,TN | Feb 2022-June 2024
Led custom design and fabrication of parametric metalwork and design options across multiple projects. Oversaw design and documentation for large-scale multifamily and civic developments in Nashville, TN. Coordinated with consultants and clients throughout all phases of the development process.

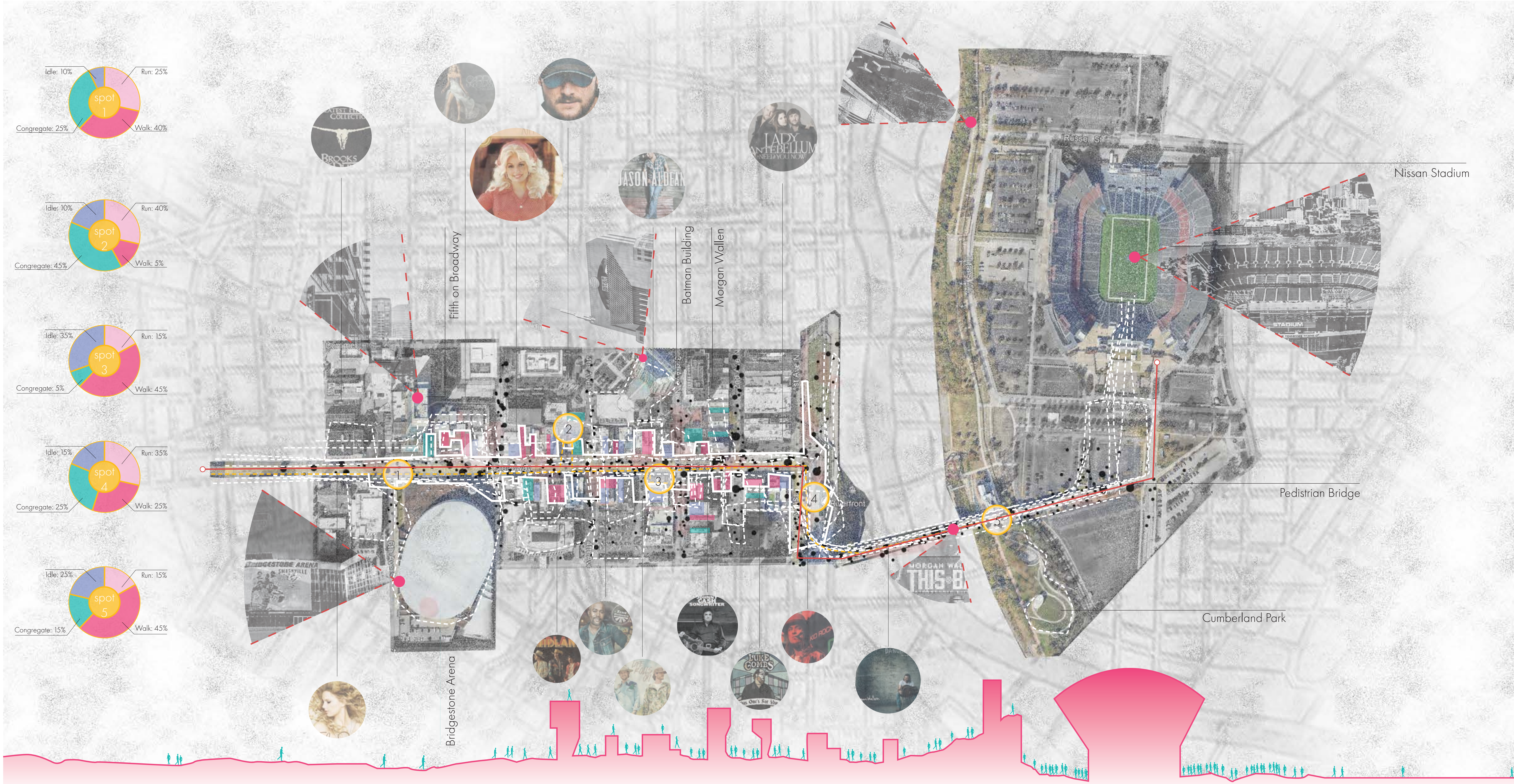
Cratus Asset Management
Contractor | New York,NY | Sep 2022-June 2024
Underwrote ground up modular home construction, RV parks acquisitions and multi-family preferred equity investment. Provided architectural services including site selection, entitlement and design development.

AECOM
Architectural Designer | Roanoke,VA | May 2021-Feb 2022
Performed data analysis on bases for Taiwan and Saudi Arabia. Developed matrices and provided architectural service to upgrade existing bases per Lockheed Martin and US Air Force standard.

Skills

Certifications & Training:
Registered Architect in the State of Georgia
A.CRE Accelerator (Core Curriculum)
Software Skills:
Python, C
Architecture Software:
Revit, AutoCAD, Rhino, Grasshopper, Adobe Suite, Bluebeam, Revisto

Table of Content



01. Walking Down Broadway (virtual immersive experience)

Class: 15-112
Instructor: Professor Mike Taylor
Partners: Individual Project

This project simulates the cultural fabric of Nashville through a first-person interactive game that blends **narrative design, real-time AI dialogue, and procedural content generation**. Players explore a dynamic, music-filled scene and engage in natural conversations with **AI-driven characters via the OpenAI API**. The system supports branching dialogue, contextual awareness, and ambient soundscapes, creating a responsive, open-ended experience. The urban environment is animated with real-time **graphics, lighting simulations, and event-driven mechanics**. Every element—from shifting street energy to layered live music—is algorithmically choreographed to reflect the city's personality. Built entirely in **Python**, the project uses **OpenCV** for **computer vision and the CMU Graphics Library for 2D animation**. It demonstrates **object-oriented design, modular architecture**, and real-time state management. A **Markov chain**–based dialogue engine generates coherent, varied responses to simulate spontaneous conversation. Scene behaviors are governed by finite state machines, showcasing **computational thinking in orchestrating emergent behaviors**. The project reflects a strong grasp of computational design, where **storytelling, interaction, and code structure converge to create a technically rich and emotionally engaging system.**

Please experience my game: **https://www.youtube.com/watch?v=h1ZbFu-W11s**

Design Research and Response

Elemental Design

1 Nashville Based Music

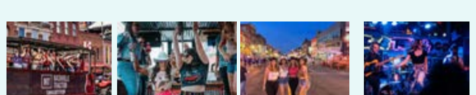
Nashville, often called "Music City," is world-renowned for its vibrant and diverse music scene. Rooted in country, blues, and rock, the city is home to legendary venues like the Grand Ole Opry and Broadway's honky-tonks, where live performances happen every night. It's a creative hub for musicians, songwriters, and producers across all genres.

2 Cityscape

Nashville's cityscape blends historic charm with modern energy. **Iconic neon-lit streets, lively music venues, and rooftop bars** line Broadway, while sleek high-rises and cultural landmarks like the **Ryman Auditorium and Bridgestone Arena** shape its skyline. The city pulses with character, creativity, and southern warmth.

3 Cultural Elements on Broadway

Broadway in Nashville is the heart of the city's nightlife—lined with legendary **honky-tonks, live music venues, and neon lights**. It's a hotspot for **bachelorette parties**, complete with **party buses**, **beer bikes**, and **dancing** in the streets, making it one of the liveliest strips in the country.



Precedent

Assassin's Creed

Assassin's Creed is a standout example of immersive game design rooted in real-world cities and history, recreating places like Renaissance Florence with **architectural and cultural precision**. Through environmental storytelling, dynamic crowds, and time-based events, it brings each setting to life. Its verticality, traversal, and mission design show how **narrative, space, and interaction** shape player experience and evoke a strong sense of **time and place.**

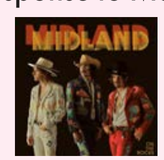
Red Dead Redemption

Red Dead Redemption is a compelling example of **immersive game design** grounded in time and place. Set in the fading American frontier, it blends **vast landscapes, historical detail, and dynamic systems** to reflect the era's tension between **wilderness and industrialization**. Its narrative and world design show how environmental **storytelling and temporal authenticity** deeply shape player experience.

Design Response

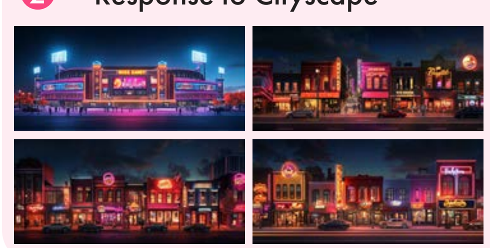
Inspired by precedents like Assassin's Creed and Red Dead Redemption, an immersive game set on Broadway in Nashville invites players to walk down the neon-lit strip, experiencing the city's vibrant nightlife in real time. Like its predecessors, the game leverages **environmental storytelling, time-based events, and character interaction** to evoke a strong sense of place. Players encounter honky-tonks spilling live music into the streets, bachelorette parties on party buses, and locals engaged in **spontaneous conversation—all powered by AI.** The design focuses on capturing the layered sounds, social energy, and spatial rhythms of Broadway, using real-world architecture, lighting shifts, and ambient noise to create a **dynamic, believable version of Music City** at night.

1 Response to Music

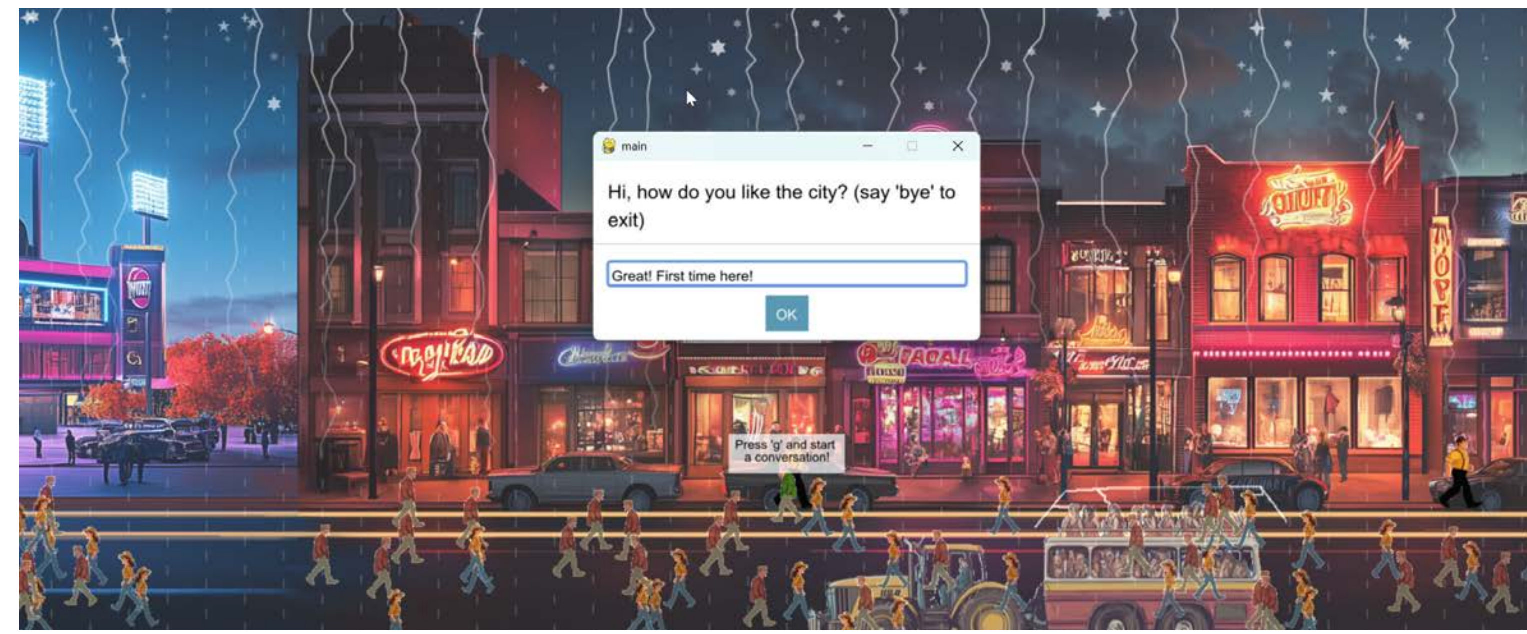


3 Response to Elements

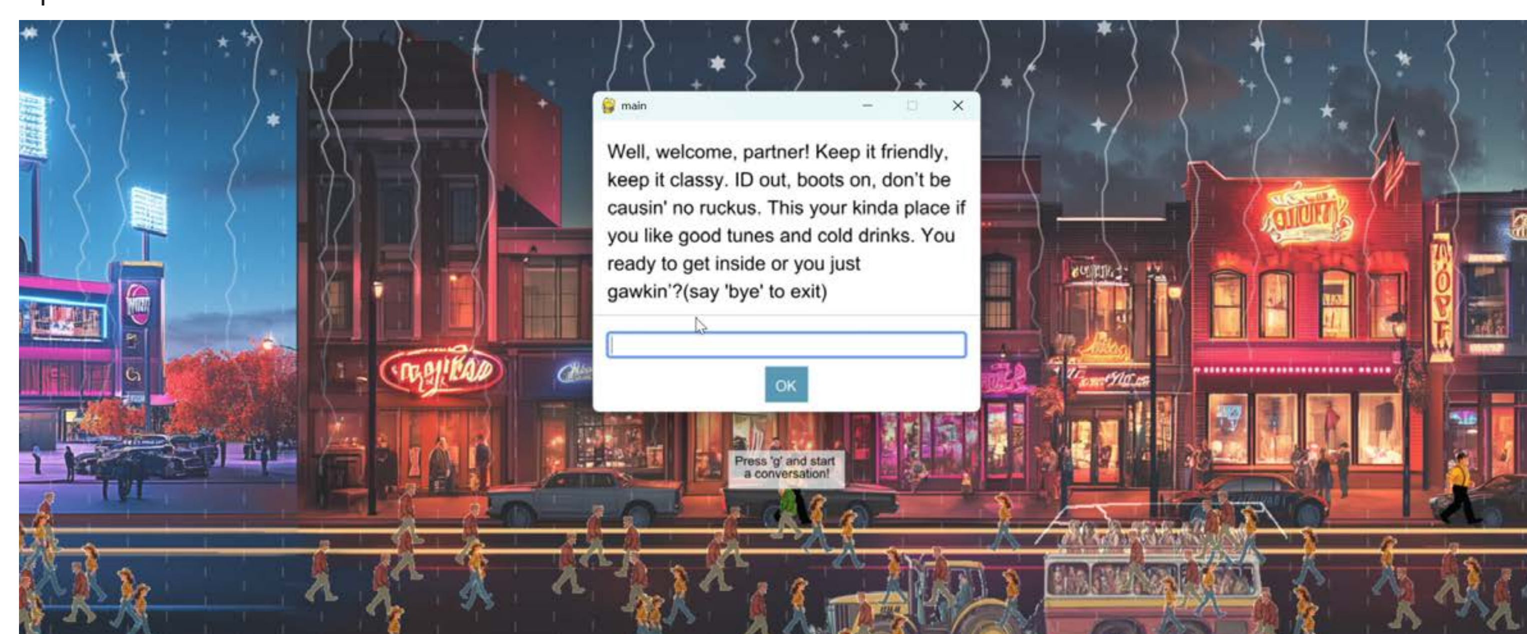
2 Response to Cityscape



Approaching NPC



OpenAI API Interface



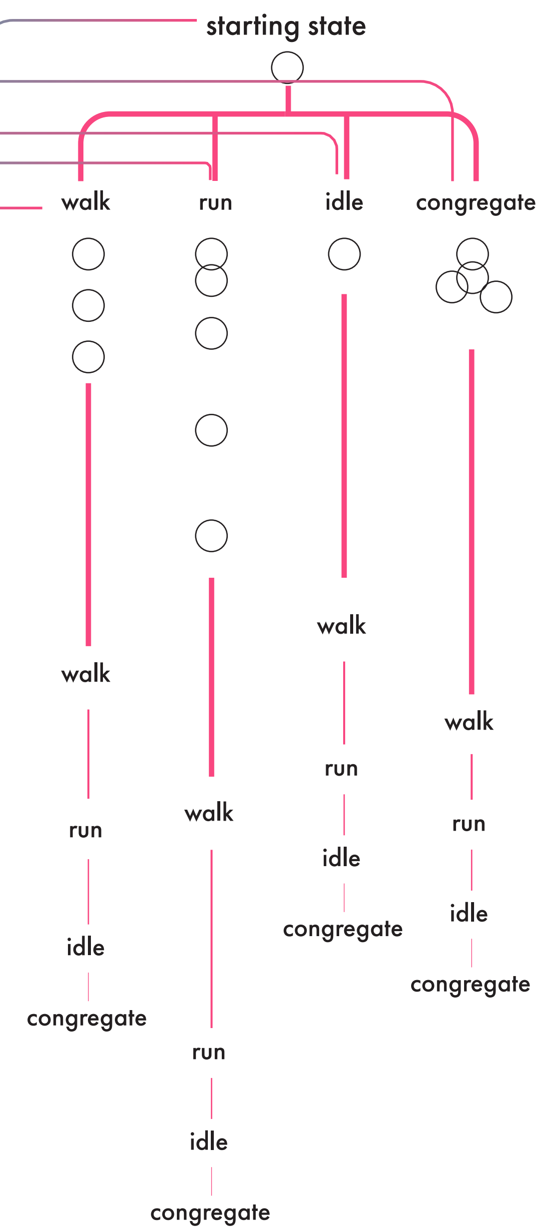
Application of Markov Chain and Computer Vision

NPC behavior is driven by a Markov chain model, where each character transitions between states—idle, walk, run, and congregate—based on probabilistic rules. Rather than scripted paths, NPCs respond dynamically to their current state and environment. For example, an NPC in the **"idle" state outside a honky-tonk may have a high probability of transitioning to "congregate"** if other NPCs are nearby, or to "walk" if the surrounding area becomes less populated. This system creates fluid, unscripted crowd behavior that mimics the spontaneous social flow of a real night on Broadway. The use of a **Markov model adds complexity and unpredictability, enhancing realism and immersion** in the virtual Nashville experience.
The game leverages **computer vision** techniques to detect bright regions within the environment and simulate the vibrant, flashing neon light atmosphere characteristic of Broadway. By analyzing **pixel intensity** in real time, the system identifies high-luminance areas—such as signs, marquees, and streetlights—and **dynamically enhances their glow and flicker to mimic the visual energy of the strip.** These light sources are further animated with varying color temperatures and pulse frequencies to create a rhythmic, immersive lighting environment. This approach not only adds visual realism but also reinforces the sensory identity of the setting, making the virtual Nashville feel alive with movement and nightlife.

Partial Code for NPC Interaction

```
def congregate(self,app):
    x = app.congregateSpot[0]
    y = app.congregateSpot[1]
    dx = 0
    dy = 0
    for people in app.peopleCluster:
        d = getDistance(x,y,people.x,people.y)
        if d<=50:
            dx = -1 if people.x>x else 1
            dy = -1 if people.y>y else 1
            people.x += dx
            people.y += dy
        else:
            self.run(app,1)
def idle(self,app):
    self.x = self.x
    self.y = self.y
def walk(self,app,factor):
    AveragePeople.run(self,app,1)
def run(self,app,factor):
    if self.x+self.dx*factor>app.width or self.x+self.dx*factor<0:
        self.dx *= -1

    answer = self.getClosest(app)
    if answer != None:
        closestPx,closestPy = answer
        if self.y>closestPy:
            self.dy = random.randint(3,5)
        else:
            self.dy = random.randint(3,5)*-1
    else:
        self.dy = 0

    if self.y+self.dy>app.height or self.y+self.dy<app.topOfStreet:
        self.dy *= -1
    self.x += self.dx*factor
    self.y += self.dy*factor
def chooseAction(self):
    num = random.randint(0,100)
    prob = dict()
    prob['walk'] = [(-5,'congregate'),(70,'walk'),(90,'run'),(100,'idle')]
    prob['run'] = [(-5,'congregate'),(50,'walk'),(90,'run'),(100,'idle')]
    prob['idle'] = [(-5,'congregate'),(70,'walk'),(90,'run'),(100,'idle')]
    prob['congregate'] = [(-5,'congregate'),(80,'walk'),(90,'run'),(100,'idle')]
    listAction = prob[self.currAction]
    for action in listAction:
        probablity, activity = action[0],action[1]
        if num<=probablity:
            self.currAction = activity
            return
```

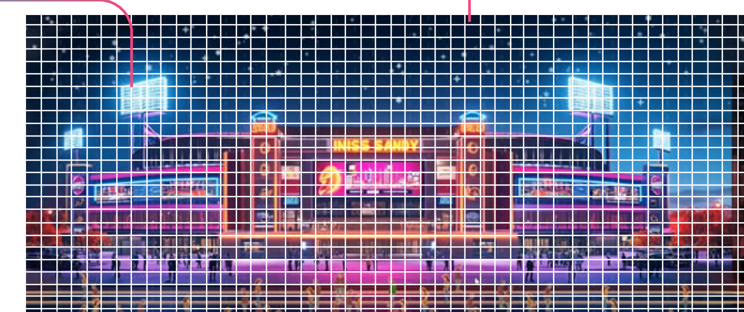
Diagram of NPC Interaction



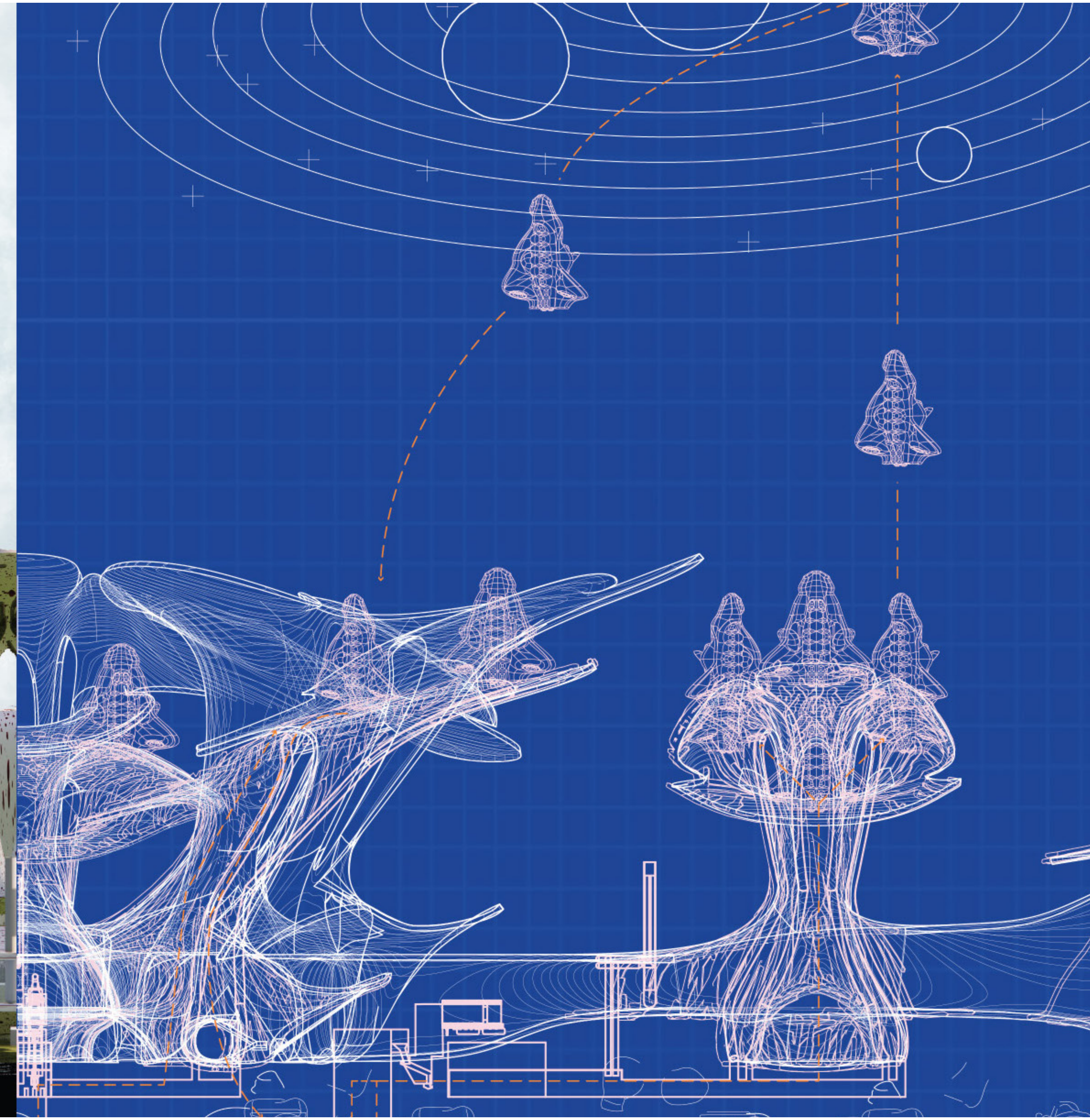
Regular Scene

```
def url_to_image(url,readFlag = cv.IMREAD_COLOR):
    resp = urlopen(url)
    image = np.asarray(bytearray(resp.read()),dtype='uint8')
    image = cv.imdecode(image,readFlag)
    return image

app.img = url_to_image(app.url)
app.img = cv.medianBlur(app.img,5)
ret,th1 = cv.threshold(app.img,150,255,cv.THRESH_BINARY)
app.coordinates = np.argwhere(th1)
app.sampleSize = len(th1)*0.05
print(f'smaplesize: {app.sampleSize}')

app.rows = 50
app.cols = app.rows*2
app.board = [[None]*app.cols for row in range(app.rows)]
app.board2 = app.board
app.setMaxShapeCount(4000)

sortCoordinate(app)
```

The scene was segmented into discrete cells, and the cumulative **RGB intensity** of each cell was computed. Leveraging the OpenCV library, an **adaptive thresholding** technique was applied to isolate **high-intensity regions**, producing a visual effect reminiscent of Broadway's **flashing neon signage.**



Computer Vision Finding Bright Spots(original pic top, Opencv enhancement pic bot)



02. Astra(Fentress Global Challenge)

Competition: Fentress 2100 Global Airport Challenge
Partner: Jonathan Liang
Award: Top 10 Finalists

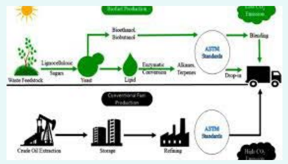
This architectural competition project envisions a **next-generation airport for the year 2100**, merging **sustainability with computational design innovation**. Developed using **Python scripting and the Grasshopper** plugin in Rhino, the design explores creative, performance-driven forms informed by **environmental data, material efficiency, and circular economy principles**.

Central to the concept is a closed-loop system that recycles urban waste and converts it into clean energy to power the airport, dramatically **reducing its carbon footprint and reliance on fossil fuels**. The design also anticipates the rise of **vertical takeoff and landing (VTOL) aircraft**, which require less runway space and allow for a more compact, multi-level terminal layout that optimizes land use. Additionally, the project **reclaims adjacent landfill sites, transforming them into thriving wildlife refuges** that restore local ecosystems, mitigate soil contamination, and support native species. A responsive skin system, optimized through **environmental simulation, regulates daylight, heat gain, and airflow across the structure**. Blending high-tech infrastructure with regenerative land use, the proposal redefines how airports can operate as ecologically integrated systems that serve both people and the planet while adapting to **future transportation and climate needs**.

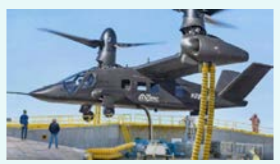
Formal and System Design

Futuristic Systems

1 Biofuel
Biofuel is a **renewable energy** made from organic waste or plants, used to power vehicles and **reduce reliance on fossil fuels.**



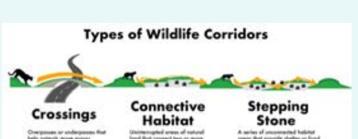
2 VTOL
A **vertical takeoff and landing (VTOL) plane** ascends and lands without a runway, allowing for **compact, flexible transport** in urban and remote areas.



3 PV (photovoltaic) Panel Farm
A PV panel farm is a **large-scale solar system** that converts sunlight into **clean electricity** for buildings and communities.

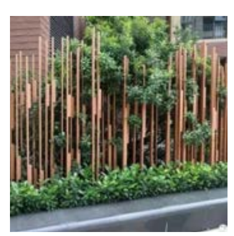
4 Hyperloop
The Hyperloop is a proposed high-speed system where **pressurized pods** travel through **low-pressure tubes** using magnetic levitation for near-**frictionless, ultra-fast** city-to-city travel.

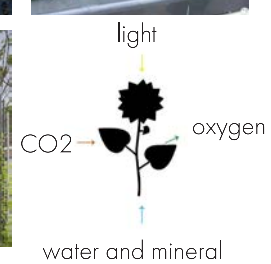
5 Wildlife Corridor
A wildlife corridor is a **natural pathway** that lets animals move safely through urban areas, **supporting biodiversity and reconnecting ecosystems.**



Design Concept

The airport is conceived as **a living plant**—a **regenerative organism** rooted in reclaimed land. Like a plant draws nutrients, filters water, and powers growth, the airport **absorbs trash as fuel, recycles water, and generates clean energy to support vertical aircraft, terminals, and shops**. This metaphor not only drives its sustainable systems but also inspires the airport's form, with **branching circulation, layered canopies, and adaptive, organic geometries** that echo the logic of **living structures.**

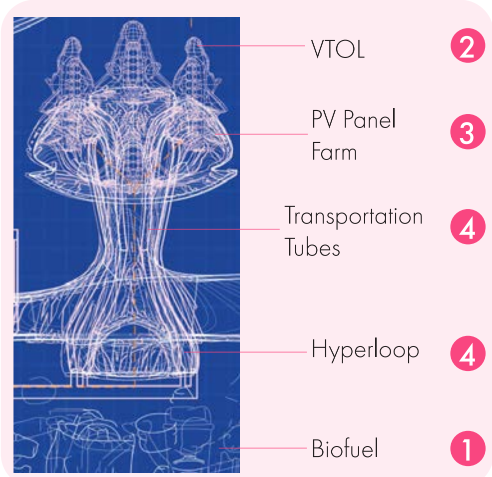


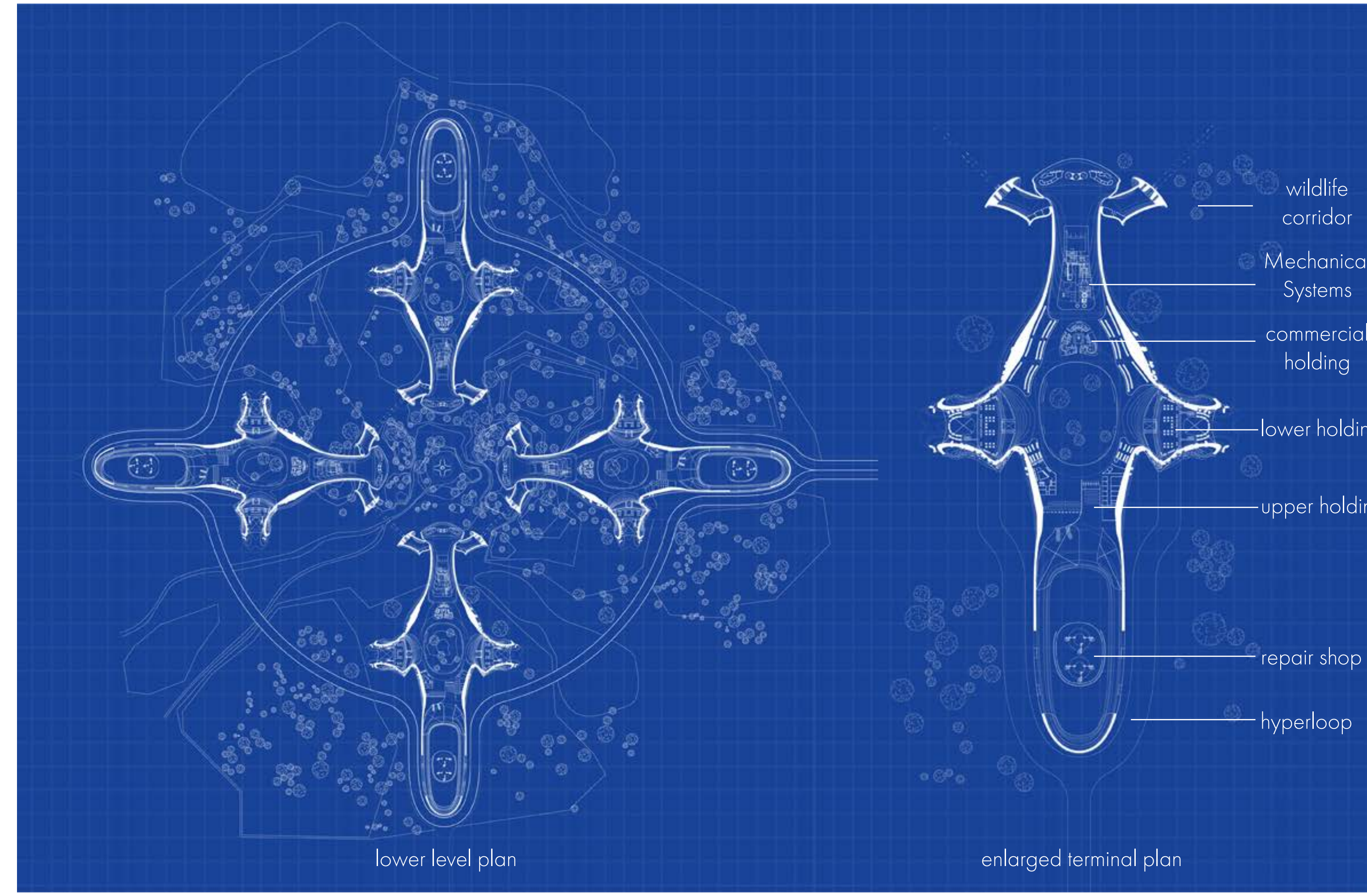


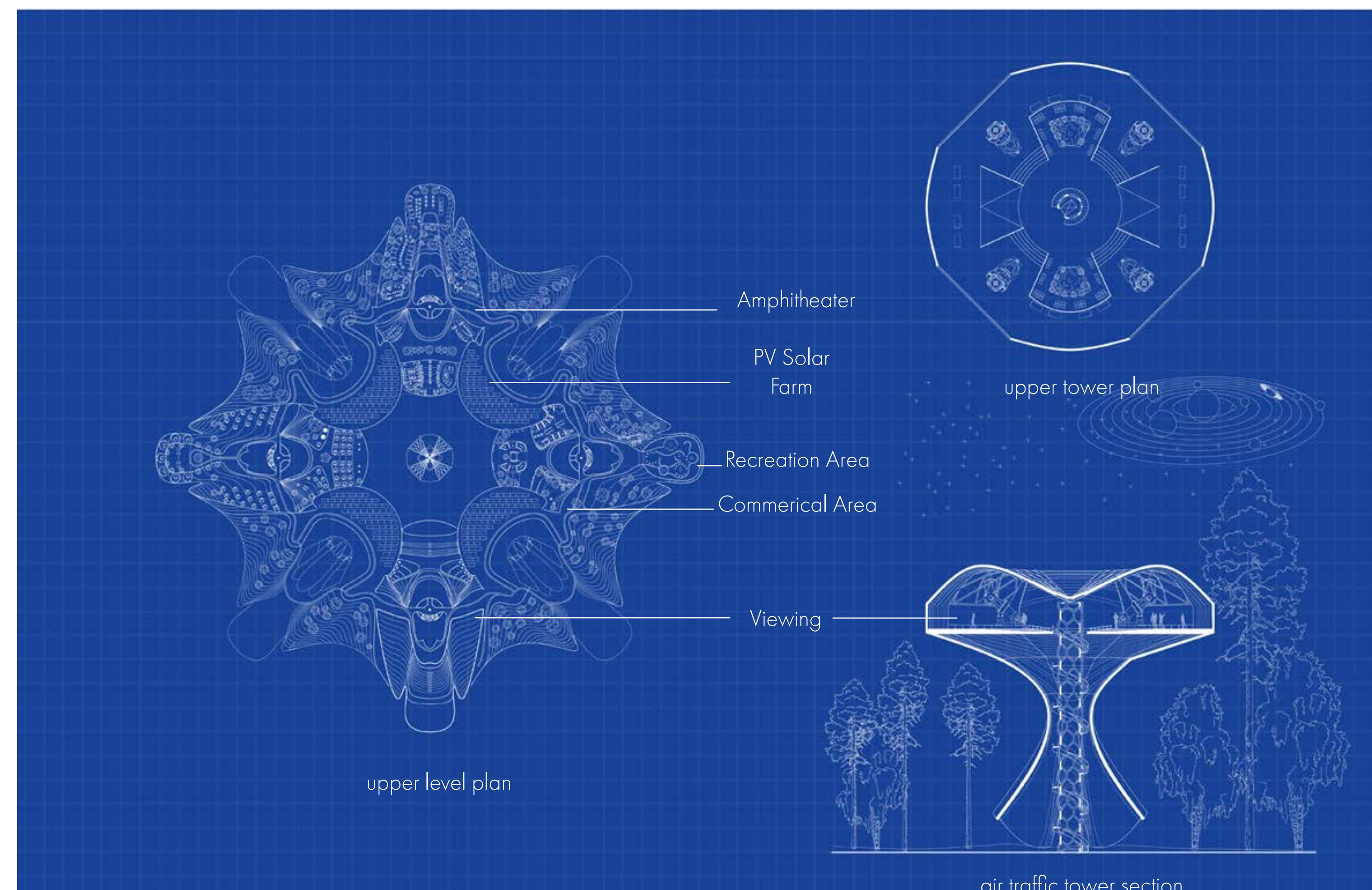
Final Design Solution

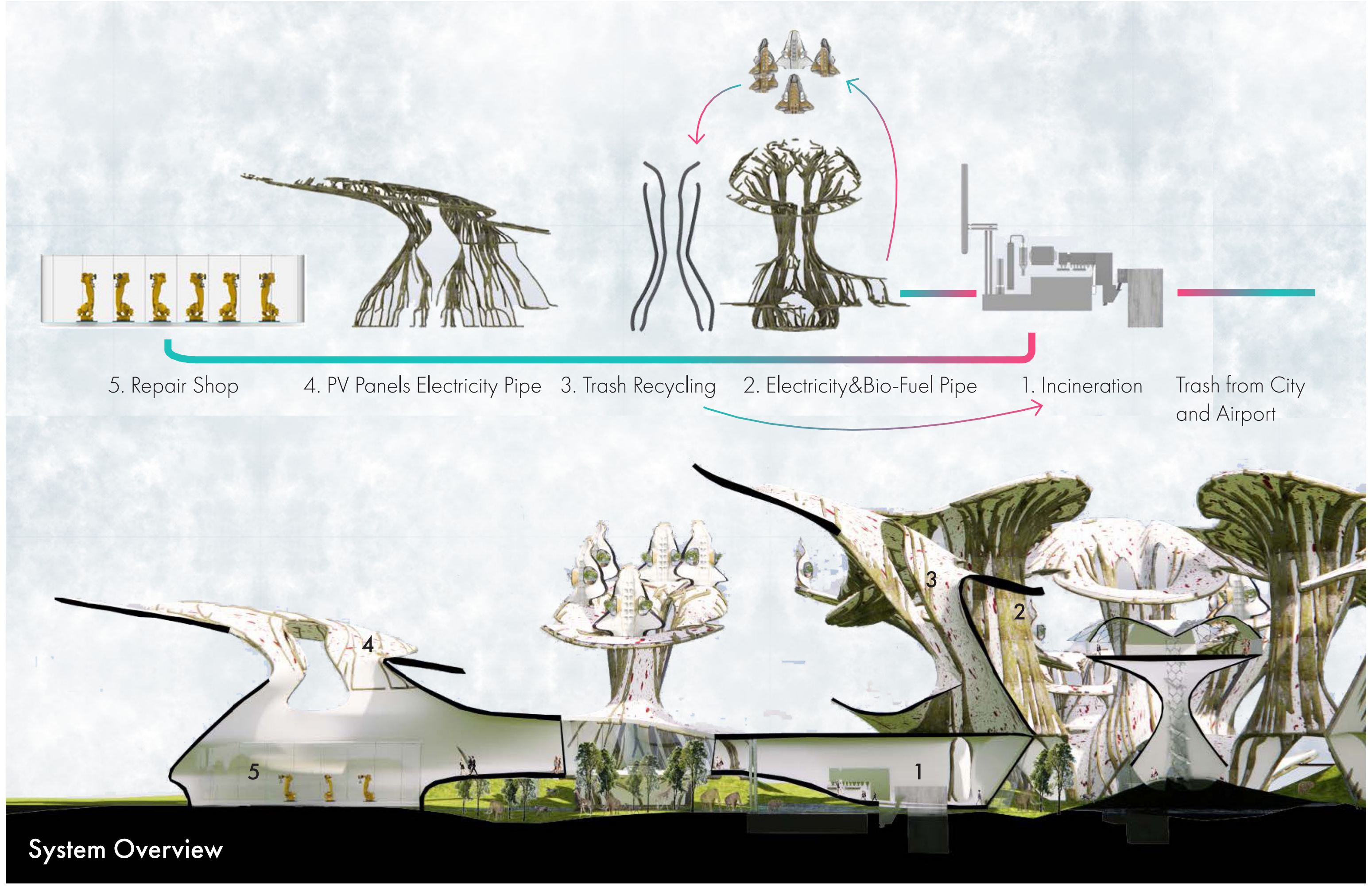
Formal Response
This airport proposal is inspired by plants, **biomimicry, and swarm behavior**—functioning like a living system that absorbs waste as energy, recycles resources, and adapts through **organic forms and intelligent, nature-driven movement.**

System Response
This future airport operates as a living system, inspired by plants, biomimicry, and swarm behavior. It **absorbs trash as biofuel, recycles water, and harvests solar energy through PV panel farms to power terminals, vertical takeoff aircraft,** and commercial spaces. Its branching form mimics plant growth, while wildlife corridors and **reclaimed landfills support surrounding ecosystems. Swarm-based algorithms guide passenger flow and autonomous vehicles**, creating a responsive, efficient, and ecologically integrated transportation hub for the year 2100.









System Overview

Overall View

Observation Tower Night View

This future airport design addresses the challenges of 2100 through **advanced technology, computational design, and sustainability**. Using Python and Grasshopper, it applies generative tools and **swarm logic** to optimize circulation and energy. Powered by **biofuel from waste, PV panels, and recycled water, it supports VTOL aircraft and terminal functions**. Inspired by **biomimicry**, the airport behaves like a living plant—**absorbing waste, purifying resources, and restoring ecosystems. Reclaimed landfills** become wildlife corridors, reconnecting habitats and enhancing biodiversity. Its **branching, data-driven form balances innovation with practicality**, reimagining the airport as a regenerative, intelligent system for people, nature, and the cities of the future.

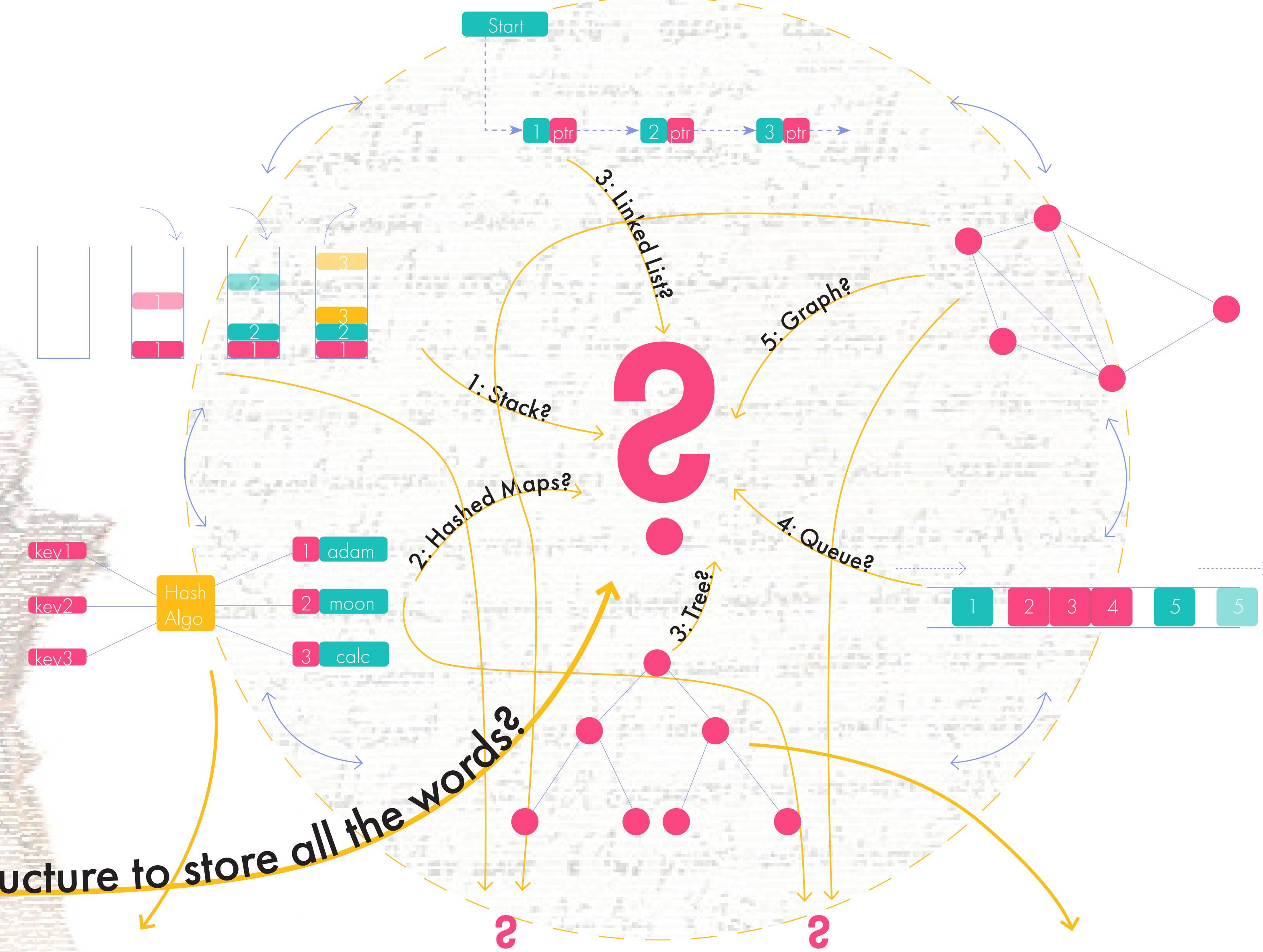
Scene 1
Synopsis:

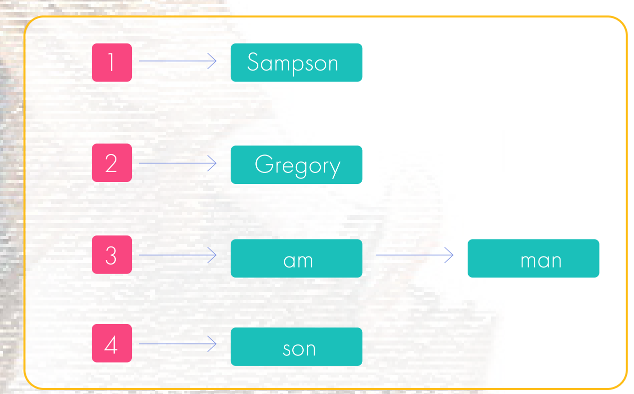
A street fight breaks out between the Montagues and the Capulets, which is broken up by the ruler of Verona, Prince Escalus. He threatens the Montagues and Capulets with death if they fight again. A melancholy Romeo enters and is questioned by his cousin Benvolio, who learns that the cause of Romeo's sadness is unrequited love.
Enter Sampson and Gregory, with swords and bucklers, of the house of Capulet.

SAMPSON **Gregory**, on my word we'll not carry coals.
GREGORY No, for then we should be colliers.
SAMPSON I mean, an weW be in choler, we'll draw.
GREGORY Ay, while you live, draw your neck out of collar.
SAMPSON I strike quickly, being moved.
GREGORY But thou art not quickly moved to strike.
SAMPSON A dog of the house of Montague moves me.
GREGORY To move is to stir, and to be valiant is to stand. Therefore if thou art moved thou runn'st away.
SAMPSON A dog of that house shall move me to stand. I will take the wall of any man or maid of Montague's.

[.........]

[how to store all the words?????]





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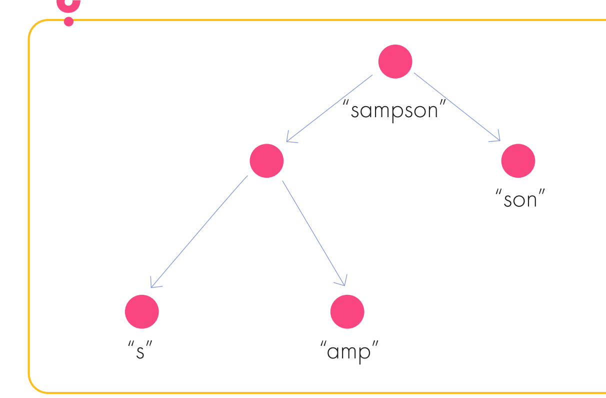

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04. Ropes(data structure design)

Class: 15-122
Instructor: Professor Mihir and Arthur
Partners: Individual Project

The simplest way to picture a string is as a single continuous line of characters — much like a ribbon with letters printed along its length. This is the way the C0 compiler stores strings: as an array of characters.
But this ribbon design has a weakness — if you want to join two ribbons, the computer must recreate the entire length from start to finish. In C0, joining two strings of lengths n and m takes O(n + m) time, which means the process slows down as the strings grow longer, much like hand-stitching two long pieces of fabric together.

The Rope project reimagines this concept. Instead of one fragile ribbon, think of a rope as a braided structure: many **short fibers (small strings) woven into a strong, branching form**. Each fiber remains independent, yet together they form a durable, adaptable whole. With this design, adding or rearranging requires working only with **select fibers rather than unraveling the entire length.** It's a structural shift — from linear fragility to modular strength — **making the rope not just a data structure, but a design for efficiency.** In C0, this rope is represented as a pointer to the following structure.

Final Design Proposal

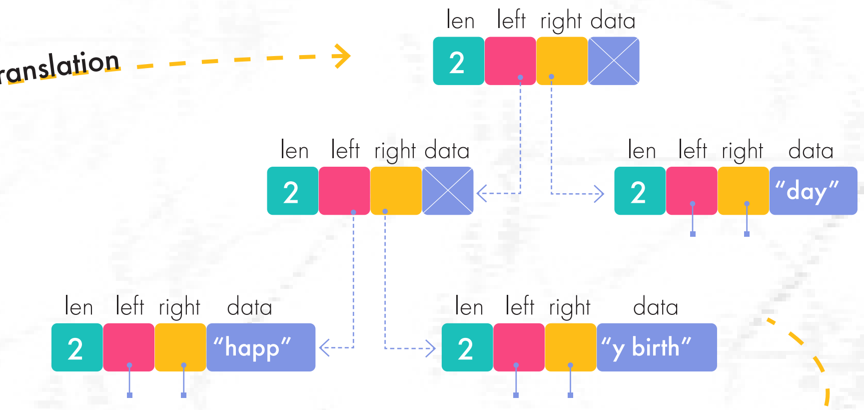
A string is usually just stored as an array of characters—that's how the C0 compiler does it too. The problem is, this setup is slow when you try to join two strings, because **combining strings of lengths n and m takes O(n + m)** time.

A rope fixes this by storing strings in a tree-like structure, which makes joining them much faster. A rope_reduce function (with a helper) takes an array of ropes, **creates a dictionary, and runs a memory-saving process on each rope**. As you go through the array, you keep earlier ropes and their parts in the dictionary so you can **reuse them when working on later ones**.

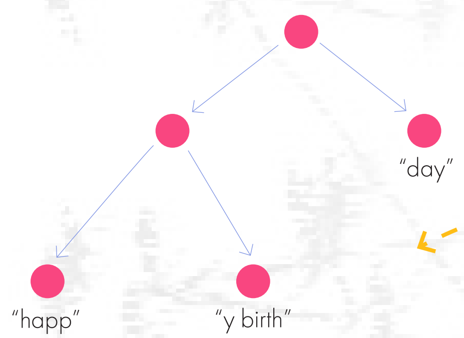
Define Type and Structure

```
typedef struct rope_node rope;
struct rope_node {
  int len;
  rope* left;
  rope* right;
  string data;
};
```

literal translation



more visual representation



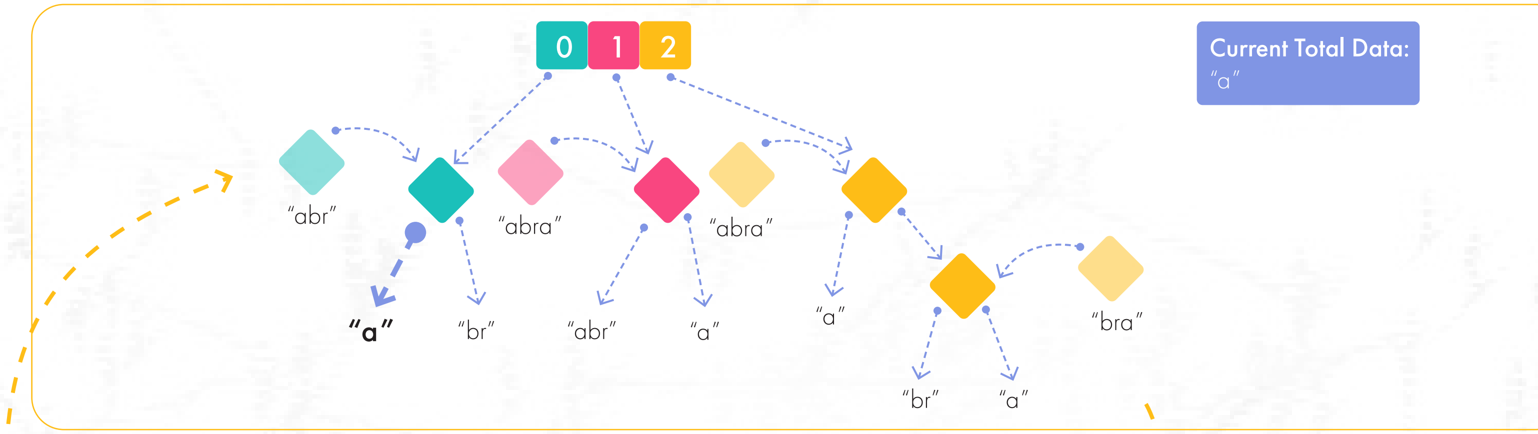
Abridged Code (picking most interesting)

```
else
{
  rope_print(R);
  printf("it entered into non_leaf \n");
  printf("the hi is %d \n",hi);
  printf("the lo is %d \n",lo);
  printf("the rope len is %d \n",R->len);
  if(lo==hi)
  {
    return NULL;
  }
  if(lo==0&&hi==R->len)
  {
    return R;
  }
  if(lo<R->left->len&&hi<=R->left->len)
  {
    return rope_sub(R->left,lo,hi);
  }
  else if(lo<R->left->len&&hi>R->left->len)
  {
    rope* left= rope_sub(R->left,lo,R->left->len);
    rope_print(left);
    rope* right=rope_sub(R->right,0,hi-R->left->len);
    rope* combined=rope_join(left,right);
    return combined;
  }
  else if(lo>=R->left->len)
  {
    return rope_sub(R->right,lo-R->left->len,hi-R->left->len);
  }
}
return R:
```

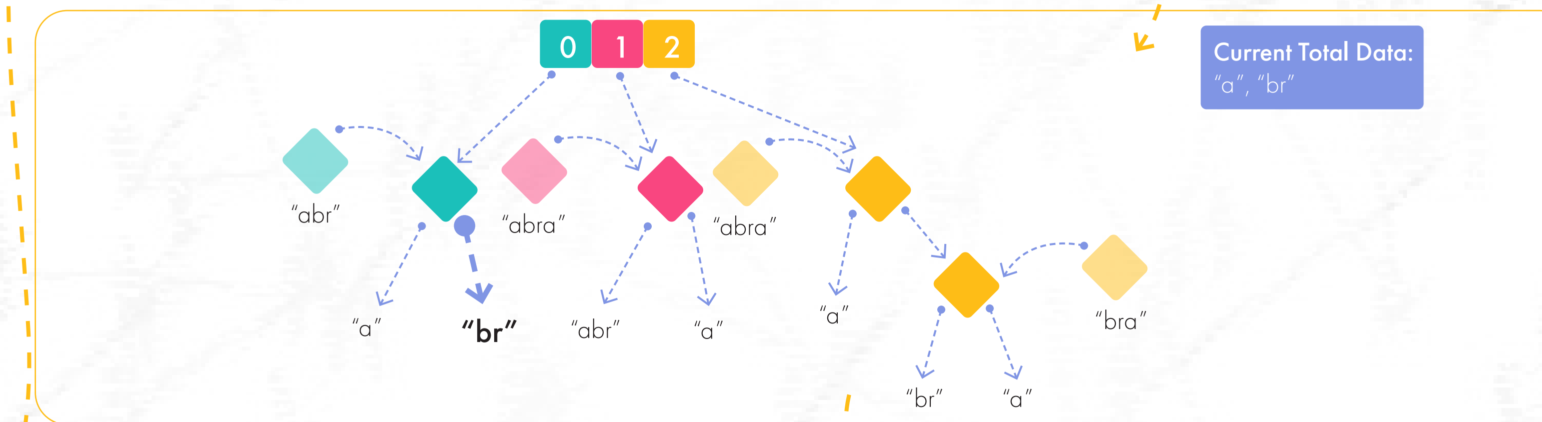
```
rope_t rope_reduce_helper(hdict_t H,entry e)
//@requires \hastag(rope*,e);
{
  if (e==NULL)
  {
    return (rope*)e;
  }
  if (is_leaf((rope*)e))
  {
    key a=entry_key(e);
    entry result=hdict_lookup(H,a);
    if (result==NULL){hdict_insert(H,e);}
    else {return (rope*)result;}
  }

  else
  // case that e is non_leaf
  {
    printf("enters non-leaf\n");
    key a=entry_key(e);
    printf("the key is: %s", *(string*)a);
    entry result=hdict_lookup(H,a);
    if (result==NULL)
    {
      hdict_insert(H,e);
      rope* temp=(rope*)e;
      temp->left=rope_reduce_helper(H,(void*)temp->left);
      temp->right=rope_reduce_helper(H,(void*)temp->right);
    }
    else {return (rope*)result;}
  }
  return (rope*)e;
}
```

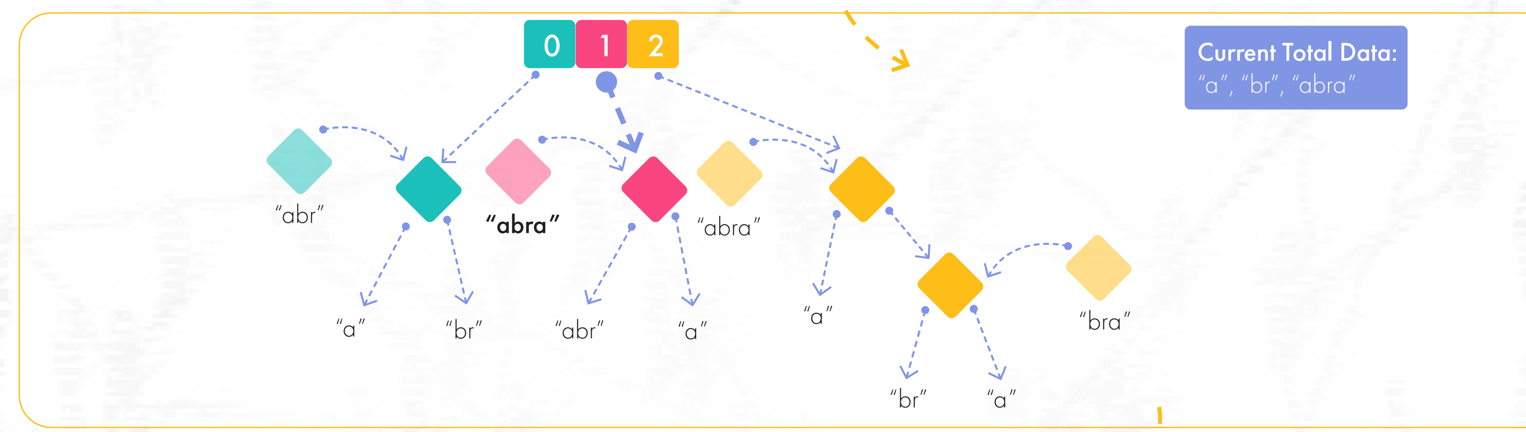
Step 1: examining data "a"



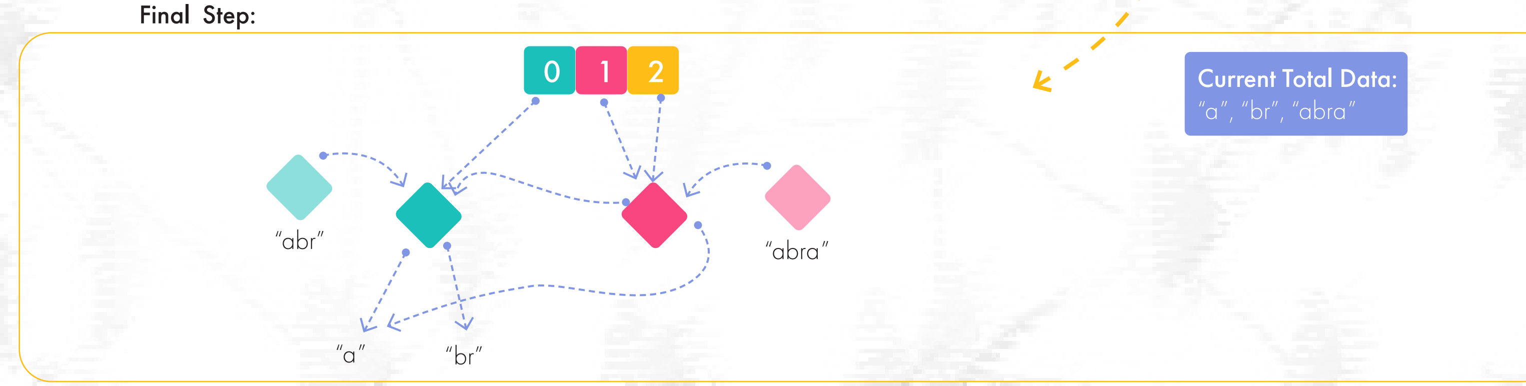
Step 2: examining data "br"



Step 3: examining data "abra"



Repeat previous steps.....
Final Step:



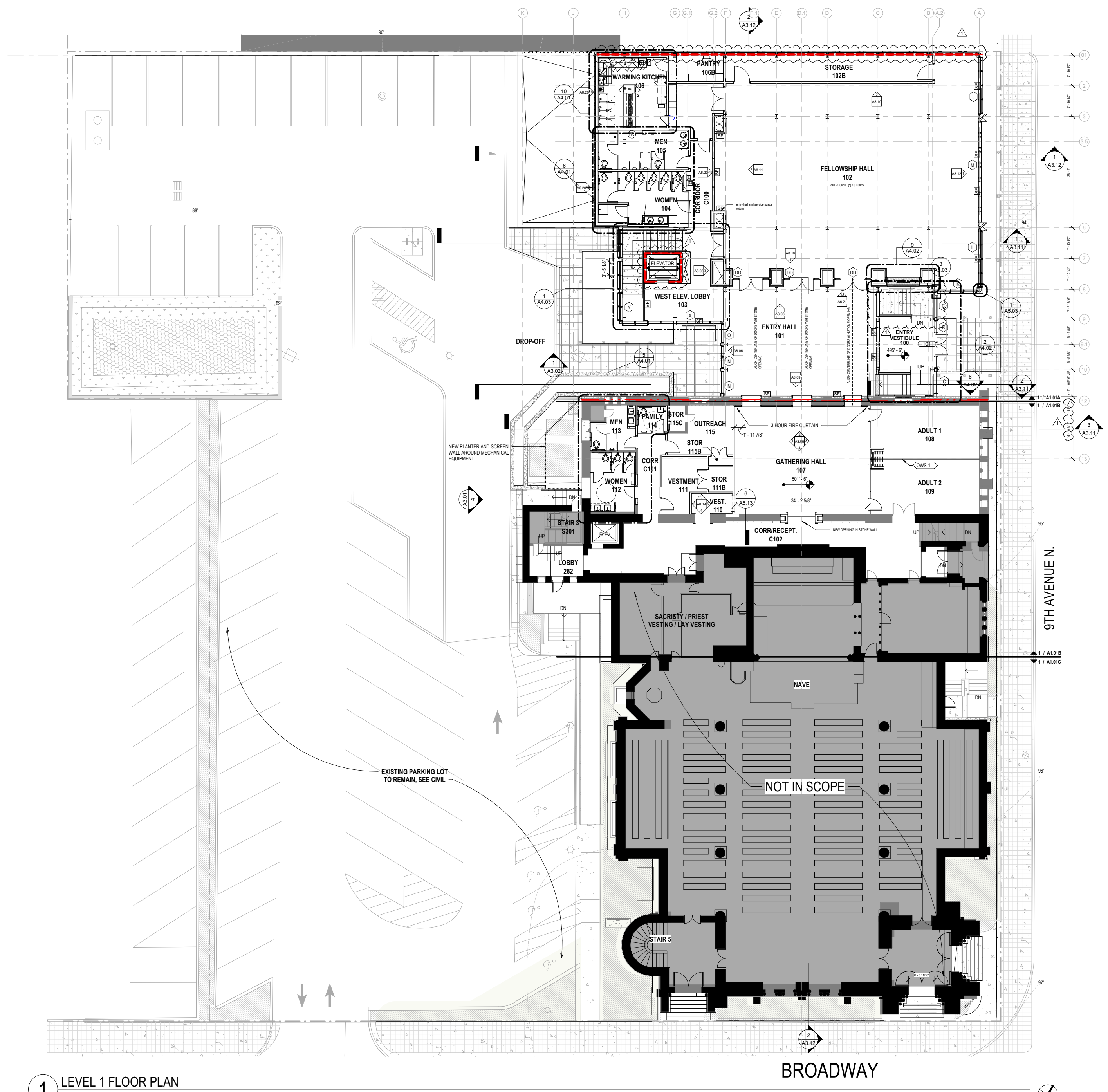
03. Christ Church Cathedral Renovation and Addition

Company: EOA Architects, PLLC
Team: 8 people team
Role: Architect leading design and fabrication of customized, waterjet cut panels. Contributed to all phases of design, client and consultant mangement

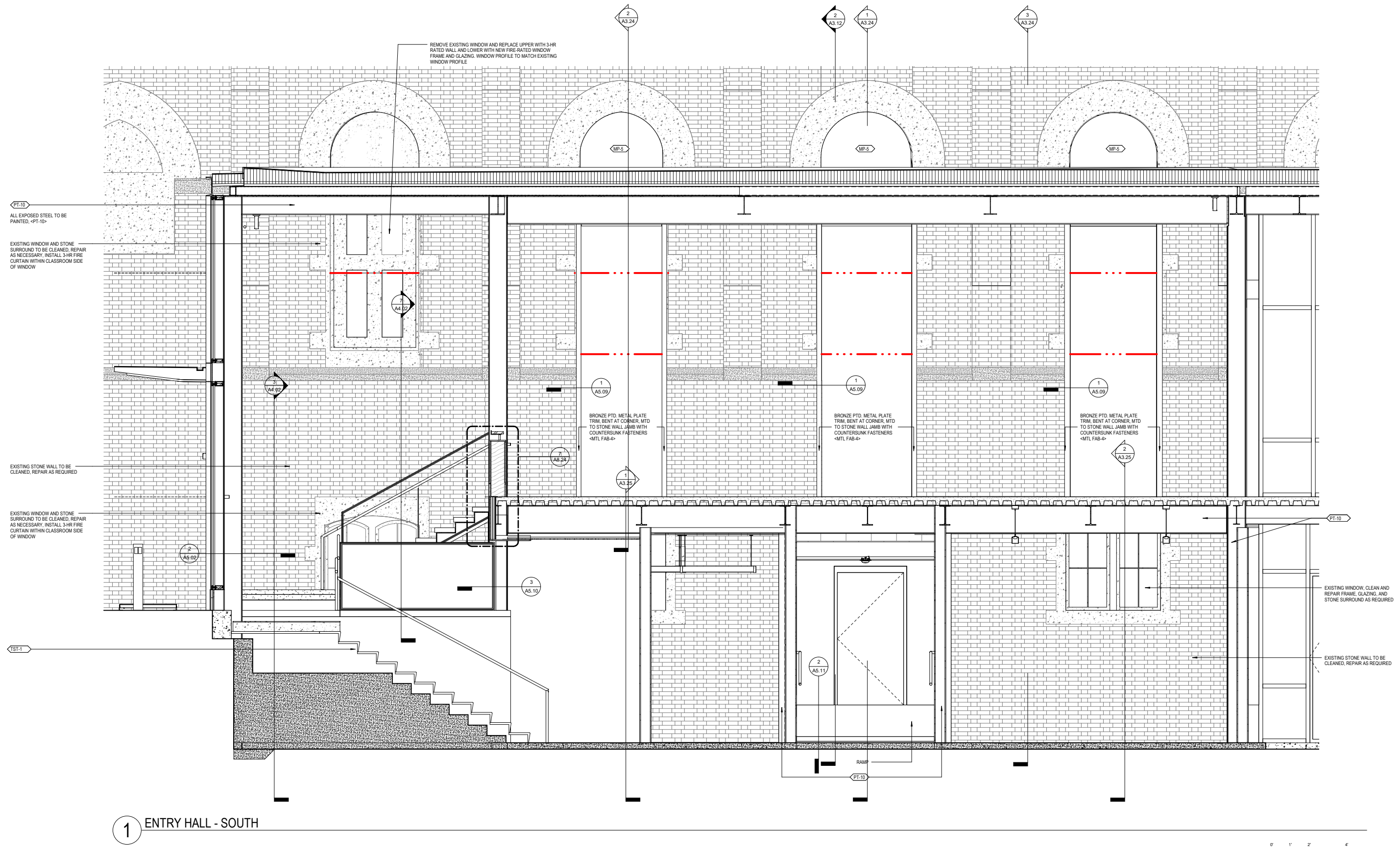
This architectural project in Nashville reimagines the historic Christ Church Cathedral—built between **1889 and 1894 in the Victorian Gothic style by Francis Hatch Kimball** and later designated as the Episcopal Diocese of Tennessee's cathedral in 1997—as a renewed **civic and spiritual hub**. Christ Church, Nashville's first Episcopal congregation founded in 1829, is notable as the diocese's "Mother Church," having helped establish numerous missions across the region. Constructed of **Sewanee sandstone and Bowling Green stone**, the cathedral features exquisite Tiffany stained-glass windows, carved woodwork, a rose window, and distinctive stone gargoyles. The renovation thoughtfully restores these elements, **reclaiming original structure and artifacts, while adding new gathering spaces and youth program areas.** The design emphasizes sustainability—reusing materials, preserving historical fabric, and enhancing energy efficiency—while remaining respectful of its architectural and cultural legacy.

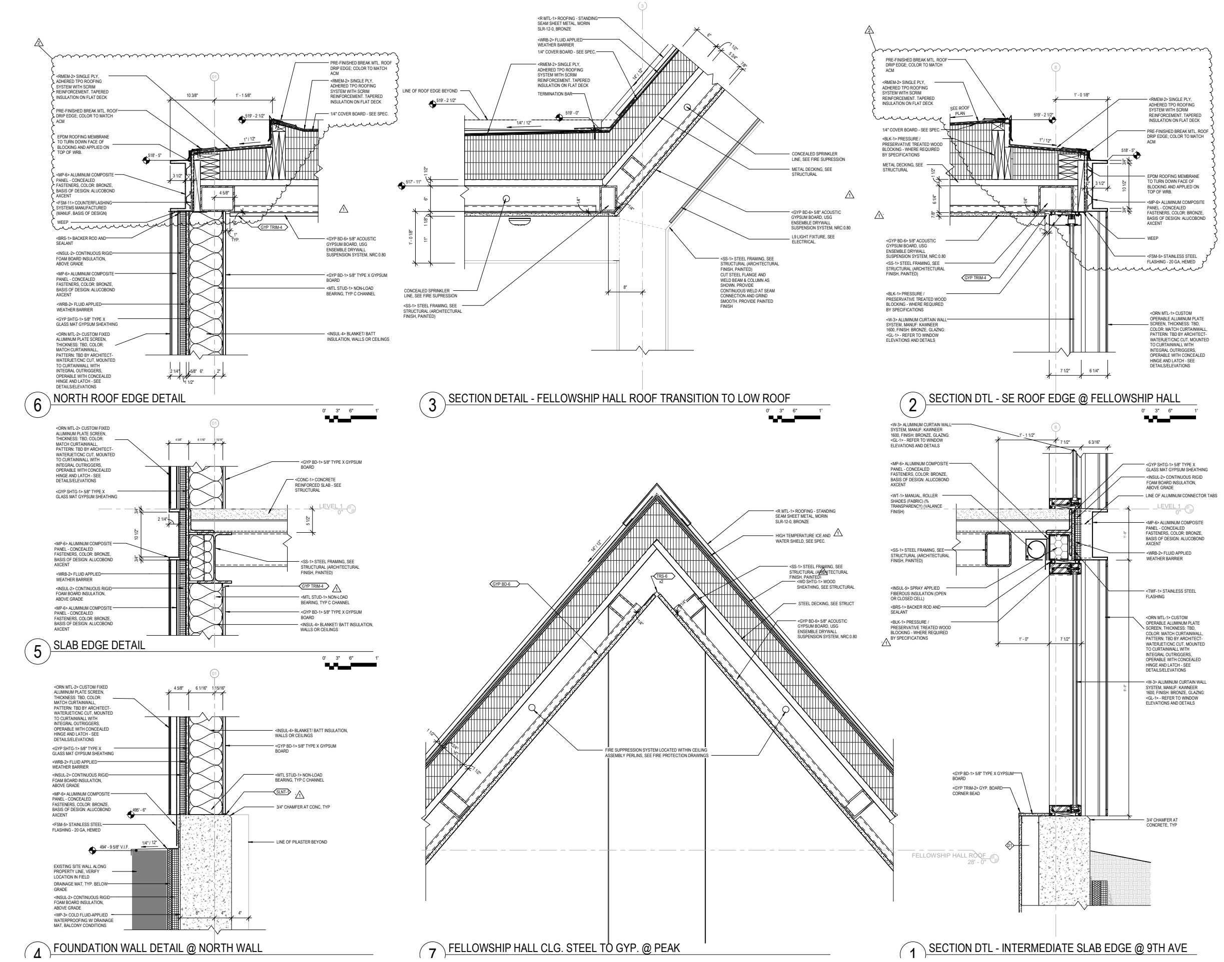
Respecting Existing and Looking to the Future

The materiality of the project bridges **historic craftsmanship and contemporary expression.** The existing church, built in the Gothic style, features locally sourced Sewanee sandstone and Bowling Green stone, with finely carved detailing and a pronounced water table that grounds the structure. The new addition draws from these defining elements—**mirroring the height, proportions, and subtle projections**—while introducing refined aluminum and steel components. Its **exterior features precision water jet–cut panels, crafted with intricate patterns that filter light** in shifting patterns throughout the day. Metaphorically, the addition becomes a **lantern—symbolizing the church as the glowing heart of its worshippers**—radiating warmth and welcome while harmonizing with the enduring strength of the original masonry.



Christ Church Cathedral as a Lantern - Filtering Light and Preaching the Gospel





Integrating New Consturction into Existing Limestone

Bronze Metal Work Details

06. Six Mile Island Residency

Class: Architecture Studio
Instructor: Professor Dana Cupkova and Heather Bizon
Partners: Individual Project

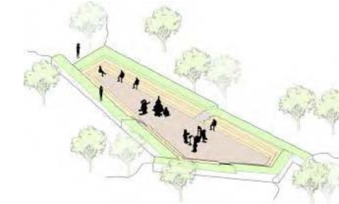
This proposal envisions a **new multi-unit residential community on Six Mile Island**, Pittsburgh, where architecture and landscape form a single, **inseparable system**. Rather than treating buildings and environment as separate domains, the project folds them into one living framework that responds dynamically to the island's **ecological cycles.**

The design is guided by a deep analysis of **wind, water, tides, and seasonal shifts, as well as the patterns of local flora and fauna**. These forces shape everything from building orientation and material selection to the arrangement of public and private outdoor spaces. Seasonal flooding and river currents inform elevated structures and **permeable landscapes**; prevailing winds and **solar paths influence façade performance and natural ventilation; migratory birds and pollinator insects guide habitat integration.**
Central to the vision is a strategy of sustainable systems woven into everyday life. **Eco-machines** recycle water through natural processes of wetland plants and microbes. **Bioswales** filter stormwater while forming ecological corridors that support biodiversity. **Productive landscapes—edible gardens, pollinator meadows, and riparian buffers**—are not added amenities but integral elements of the residential fabric.
The result is a **new typology of sustainable living**, where residents engage with their environment not as passive observers, but as participants in an evolving ecological system. Six Mile Island becomes both home and habitat: a **prototype for resilient riverine urbanism** in Pittsburgh and beyond.

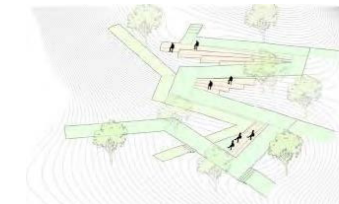
Translating Analysis into Architecture

The ecological analysis informs the architectural expression. Building orientation follows **prevailing winds and solar angles**, reducing mechanical loads while maximizing daylight and ventilation. Elevated foundations respond to **flooding and shifting river levels**, while porous ground floors dissolve into **bioswales and wetland gardens that filter runoff.** Roofscapes act as ecological membranes—layered with solar arrays, rainwater harvesting systems, and pollinator habitats—**blurring boundaries between infrastructure and landscape**. Units are staggered to create cross-ventilation corridors and shaded terraces, while shared courtyards double as **stormwater catchment zones and social gathering spaces**. The architecture is not an object placed on the island, but a responsive system translating **natural forces into spatial, structural, and communal form.**

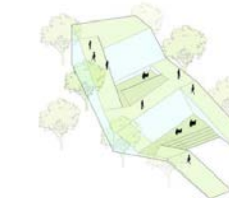
1. Indented Outdoor Space

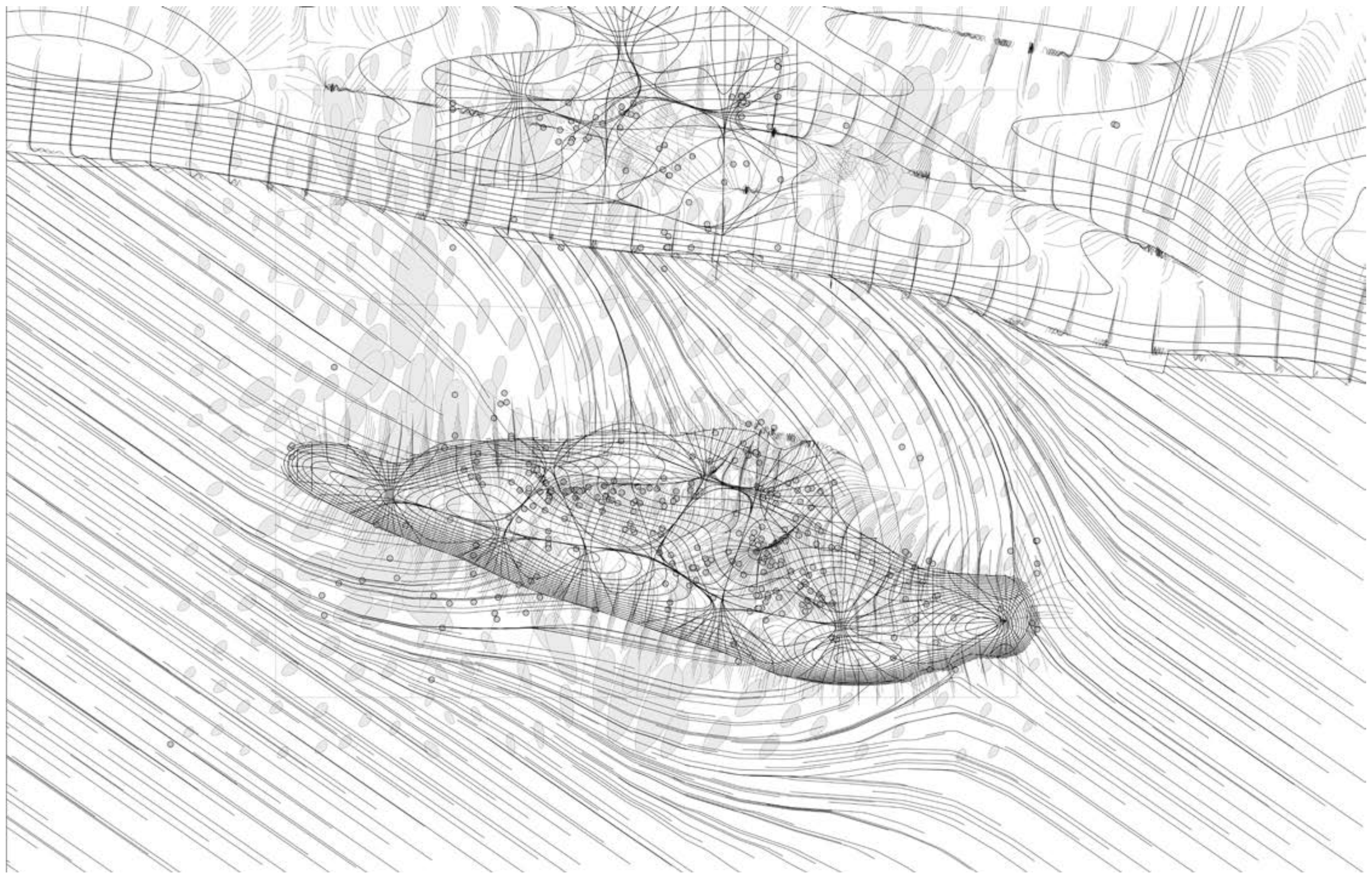


2. Serpentine Walkway



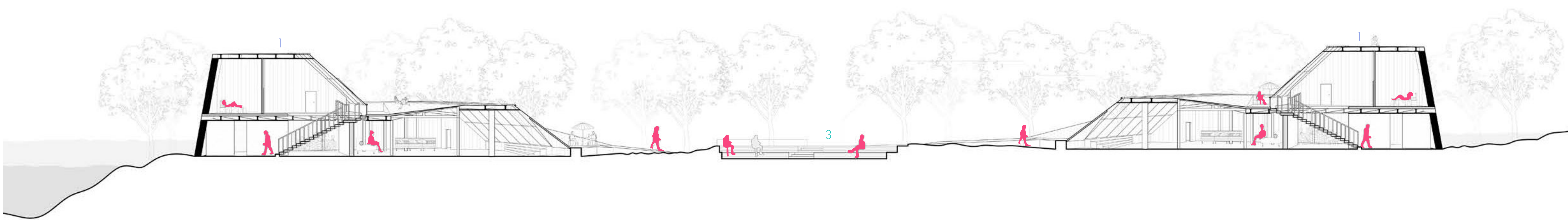
3. Residency





Analytical Plan

Architecture from Analysis

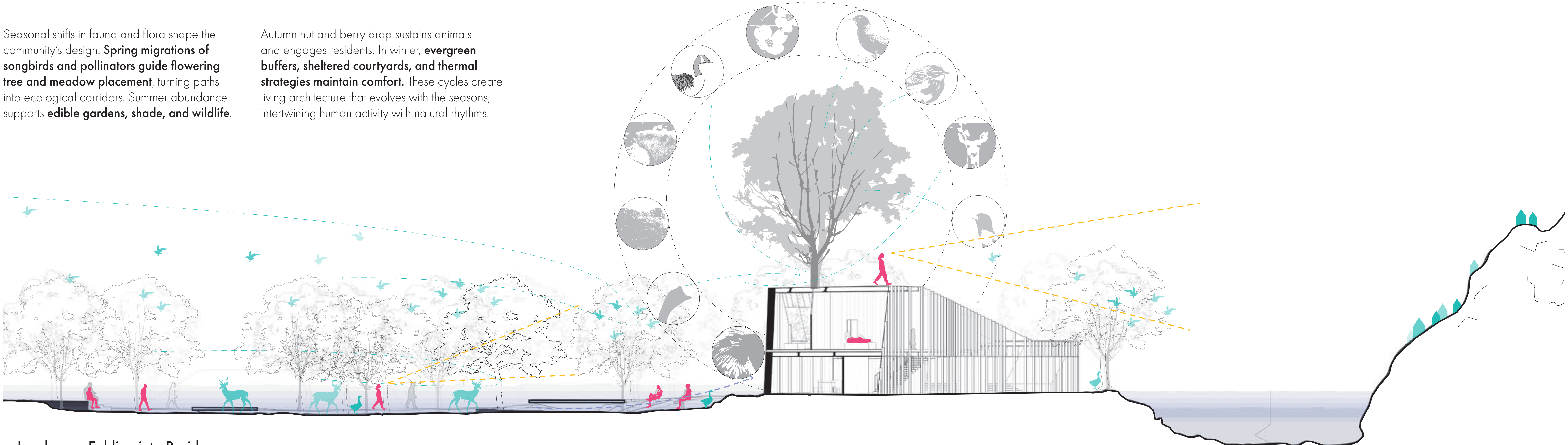


Landscape Folding into Residency

Seasonal Fauna and Flora

Seasonal shifts in fauna and flora shape the community's design. **Spring migrations of songbirds and pollinators guide flowering tree and meadow placement**, turning paths into ecological corridors. Summer abundance supports **edible gardens, shade, and wildlife**.

Autumn nut and berry drop sustains animals and engages residents. In winter, **evergreen buffers, sheltered courtyards, and thermal strategies maintain comfort.** These cycles create living architecture that evolves with the seasons, intertwining human activity with natural rhythms.



Landscape Folding into Residency

View 1: Inside of the Landscape

View 2: Top of Residency

06. Vanderbilt University Residency and Event Center(LEED Plat)

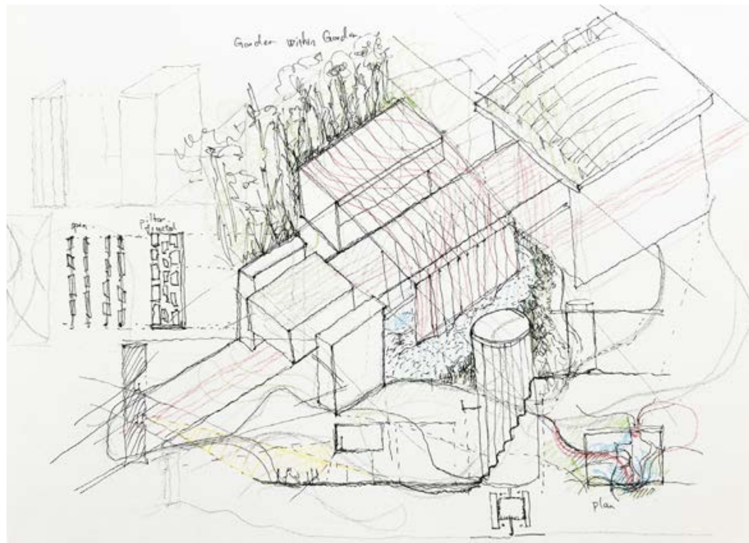
Company: EOA Architects, PLLC
Team: 5 people team
Role: As project architect, I produced all renderings, contributed to all drawing deliverables, and participated in OAC and coordination meetings.

EOA Architects provided architectural services from feasibility study through design development for a **new university residence and event center**, along with its surrounding garden. Vanderbilt University, one of the most historic and prestigious research universities in the nation, is renowned for its **rigorous academics, distinguished faculty, and picturesque campus.**

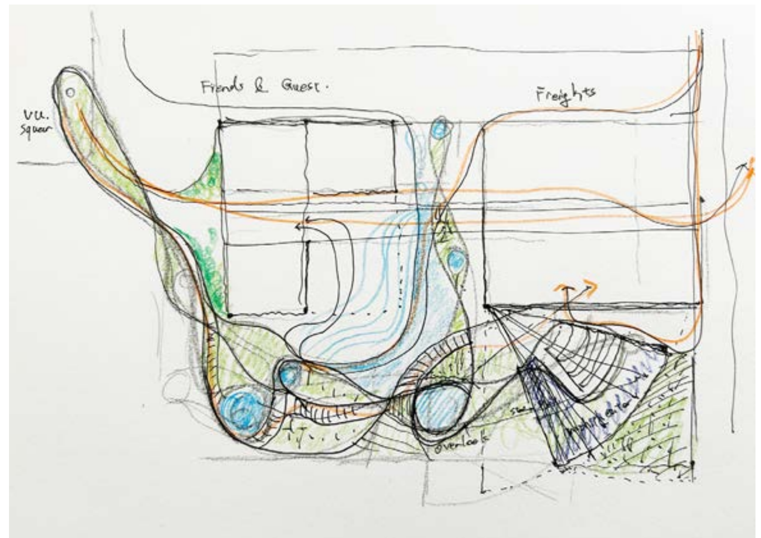
The new residential design aims to embody the university's rich history and values.
The site and building embody the concept of a **"Garden within a Garden."** The expansive garden is open to guests and students, offering both a living and educational experience. Visitors can engage with the **LEED Platinum** design, experiencing the seamless **integration of nature, sustainability, and education.** The main residence features its own arbor and water garden, providing the host with **serene, reflective moments amidst their busy lives.**
To further connect architecture with meaning, the residence features an **entry portal—symbolically called the Pillar of Studies**—representing the transition from the vibrant energy of student life to a more reflective, restful, and didactic environment. Targeting LEED Platinum certification, the project underscores the university's commitment to sustainability.

Pillar of Study

Vanderbilt University is one of the **most historic and prestigious research universities** in the nation. Renowned for its rigorous academics, distinguished faculty, and picturesque campus, the new residential design aims to embody the university's rich history and values. To achieve this, the residence features an **entry portal that serves as a symbolic 'Pillar of Studies,'** representing the **transition from the vibrant energy of student life to a more reflective, restful, and didactic environment.**

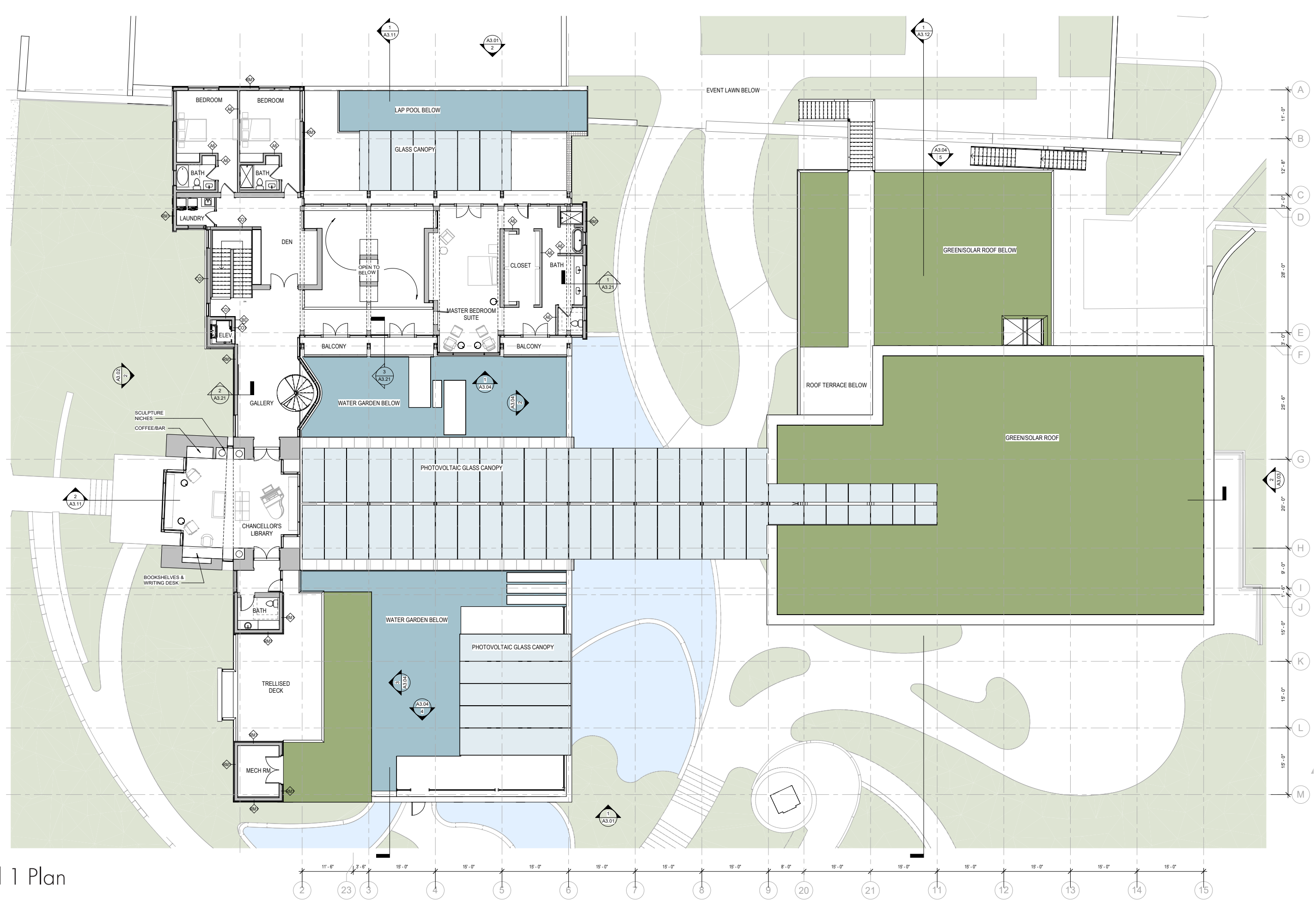


Design Comic - Garden within a Garden



Design Comic - Site Strategy

Entry Portal

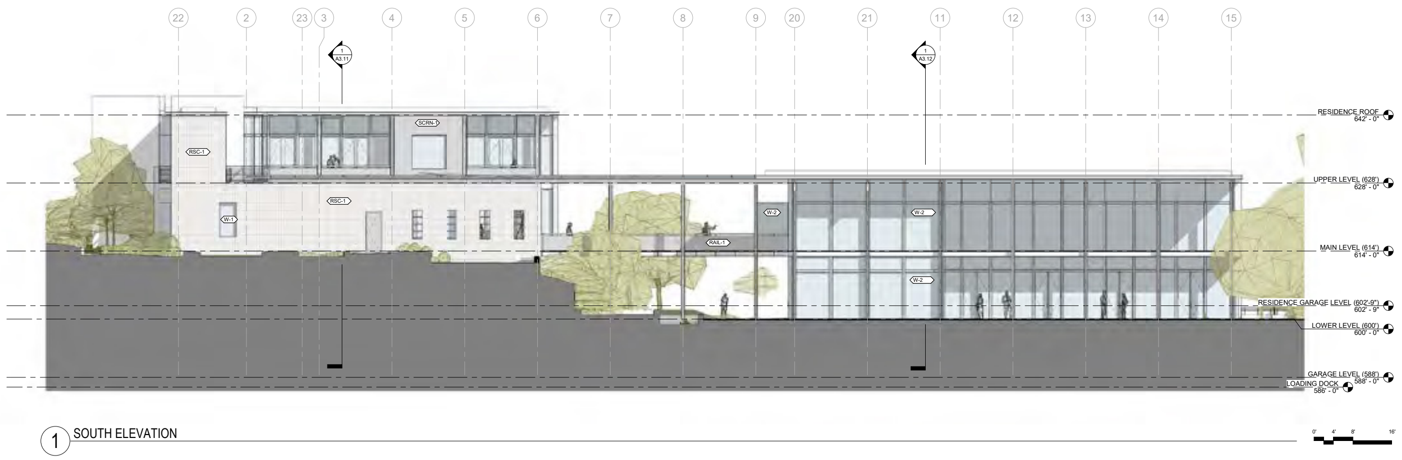


Level 1 Plan

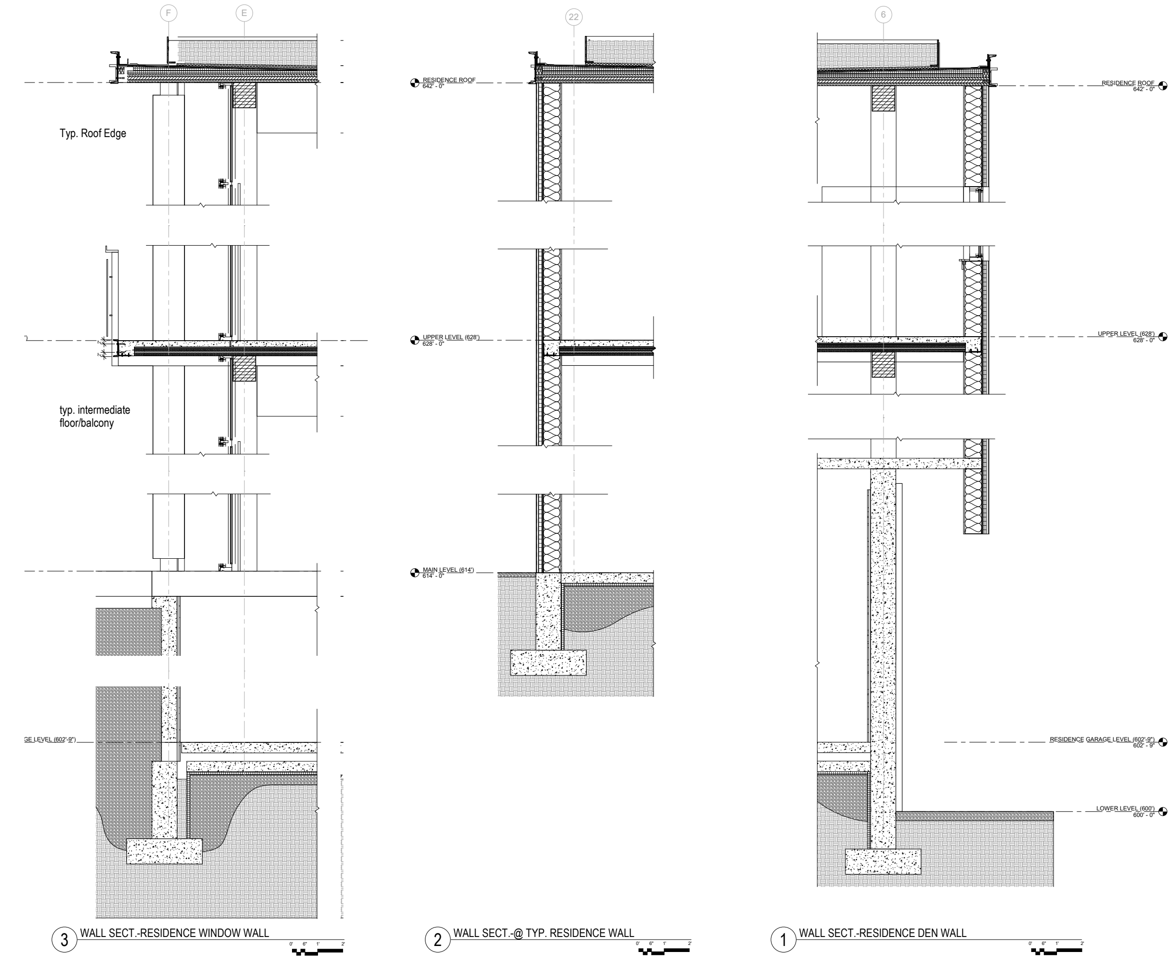
Arbor in Inner Garden

CLT and Steel Structure Expression

A building that combines **Cross-Laminated Timber (CLT)** with a **steel structure** offers a balance of sustainability, strength, and design flexibility. CLT, being a r**enewable and low-carbon material**, significantly reduces the building's environmental footprint while providing **excellent thermal performance and a warm, natural aesthetic.** Steel, on the other hand, offers exceptional tensile strength, allowing for **longer spans, slimmer profiles,** and greater open floor plans without compromising structural integrity. Together, they create a hybrid system that optimizes material use—steel handling **high loads and lateral forces,** while CLT delivers lightweight, prefabricated panels for rapid assembly. This synergy results in a high-performance building that is **efficient to construct, adaptable in design, and resilient over its lifespan.**



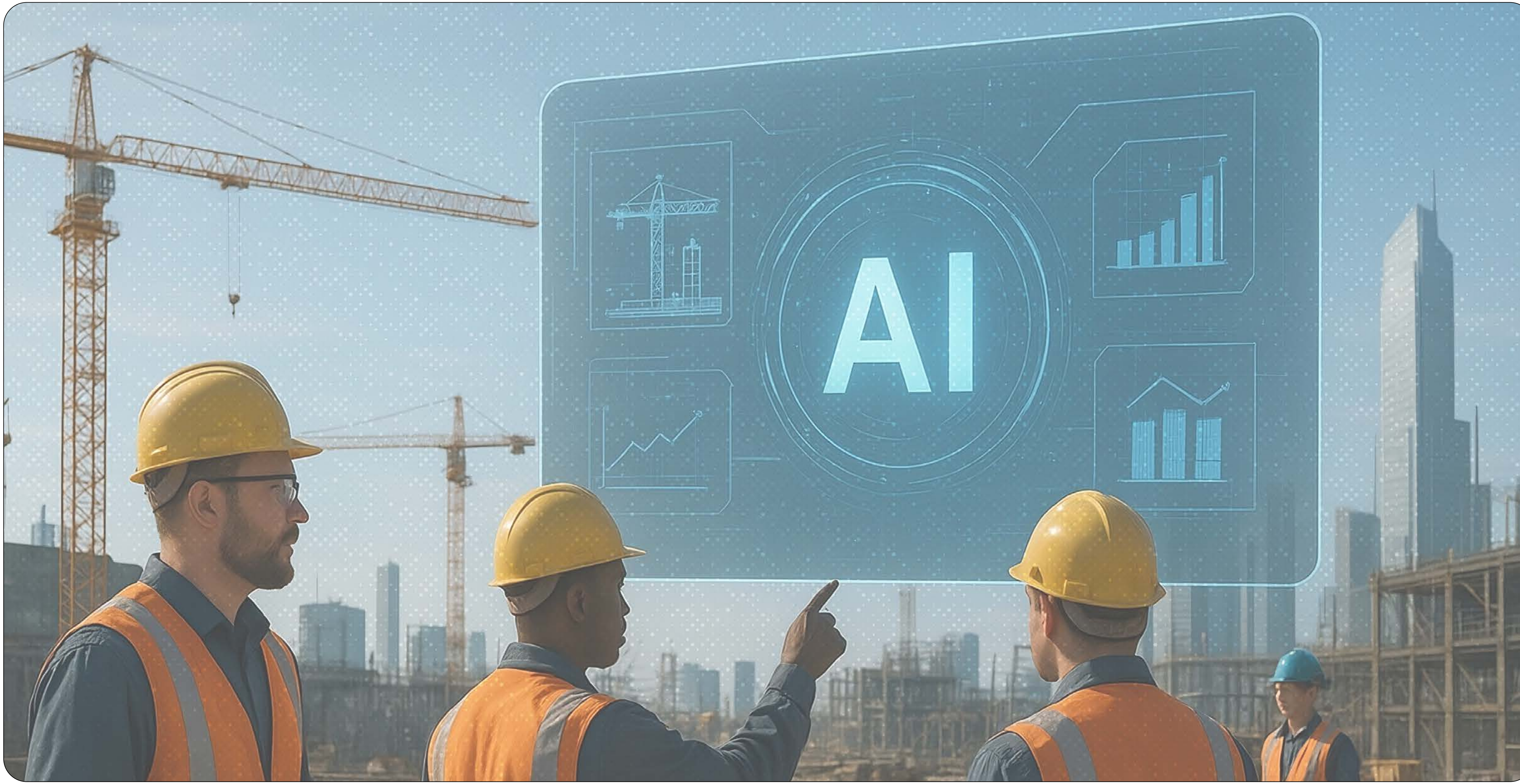
Portal to Event Center Section



General CLT Wall Section

Garden View

Reception Hall



05. Post OSHA

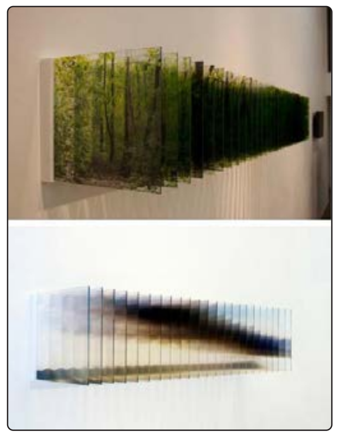
Team: Individual Project
Role: Designed the user interface for a next-generation construction application that leverages computer vision and machine learning to enhance safety, optimize project timelines, and reduce costs.

This project involves the design of a **user interface** for a cutting-edge construction application that integrates **computer vision and machine learning.** The platform analyzes real-time site data to **detect hazards, track progress, and optimize resource allocation**, ensuring safer, faster, and more cost-efficient construction. For contractors, the technology **minimizes downtime, streamlines workflows, and prevents costly errors**. Design professionals gain data-backed insights to **improve coordination, design accuracy, and construction sequencing**. Regulatory agencies such as OSHA benefit from automated compliance monitoring, **reducing workplace incidents and ensuring safety standards** are met.

With the United States facing a **shortage of millions of homes,** this technology offers a pathway to accelerate construction while maintaining high safety and quality standards—**bridging the housing gap without compromising worker welfare.**

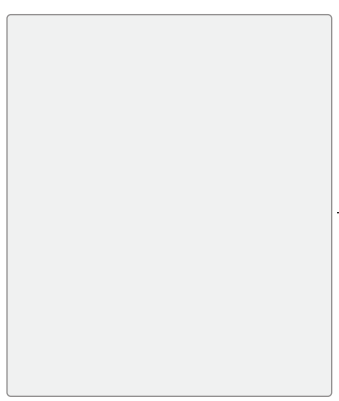
Case Study: 715 Merritt Ave, Nashville, TN

At 715 Merritt Ave, two major issues occurred during construction: the **sidewalk was poured at the wrong elevation**, disrupting ADA access, and the grand stairs were built **differently** from the **design intent**. Both mistakes resulted in significant costs and overtime for all parties involved.

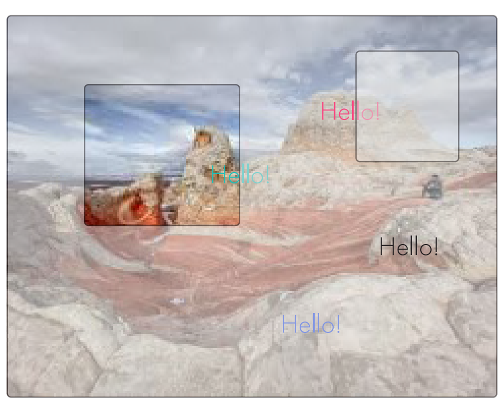


Design Concept

Glass is a unique design material—strong yet fragile, **transparent yet reflective**. Its ability to transmit, refract, and diffuse light transforms spaces with **luminosity and openness.** Designers use clarity, translucency, and color to shape environments. Artists like Dale Chihuly and Olafur Eliasson reveal glass's **poetic potential,** merging craftsmanship, technology, and vision.



In Use



Applying Physicality to Interface Deesign

In AI app interfaces, the **physicality of glass** inspires experiences that feel lightweight yet layered. **Transparency and reflectivity** translate into semi-transparent panels, blurred backgrounds, and subtle reflections, **adding dimensionality** without clutter. This aesthetic conveys **clarity and trust**—users can "see through" the system—while balancing elegance with **robust, intuitive usability**.

Governmental Point of View:

1. For construction work at elevations of 6 feet or more, **OSHA mandates** the use of: **guard-rail systems, safety net systems, or personal fall arrest systems.**

Contractor's Point of View:

1. Concrete work exceeded the specified elevation and required rework, resulting in an additional **cost of $350,000.**
2. Time was wasted constructing an incorrect sidewalk, which had to be **demolished and rebuilt.**
3. Sidewalk errors caused project delays, impacting both **schedule and profitability.**

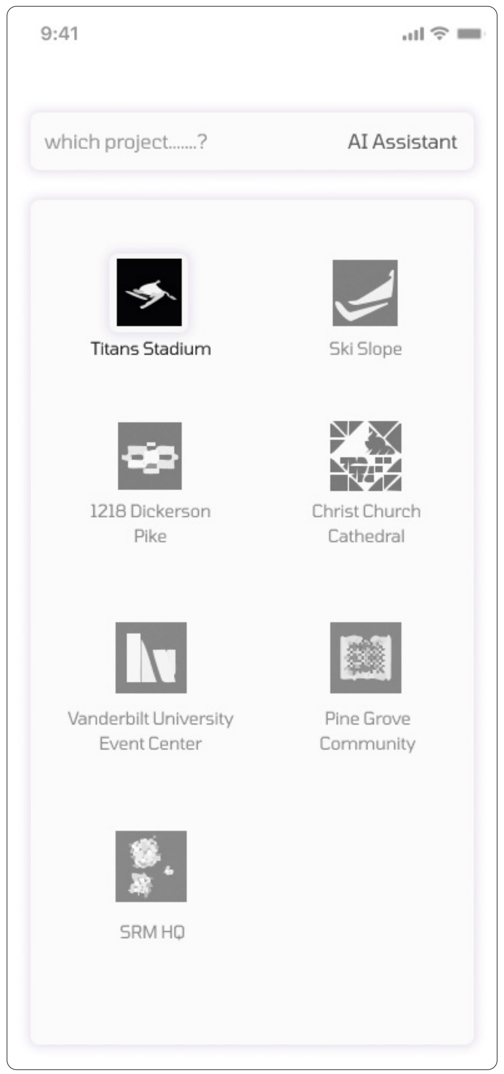
Developer Point of View:

1. **Arbitration** on who would foot the bill. Technically, it is the contractor's fault, but most of the time, contractor would negotiate with developer for a compromise.

Design Professional Point of View:

1. Overwhelmed by **endless paperwork—RFIs, change directives, change orders, and coordination tasks.**
2. Draining time, resources, and energy, leading to **significant costs and constant frustration.**

Choosing Projects

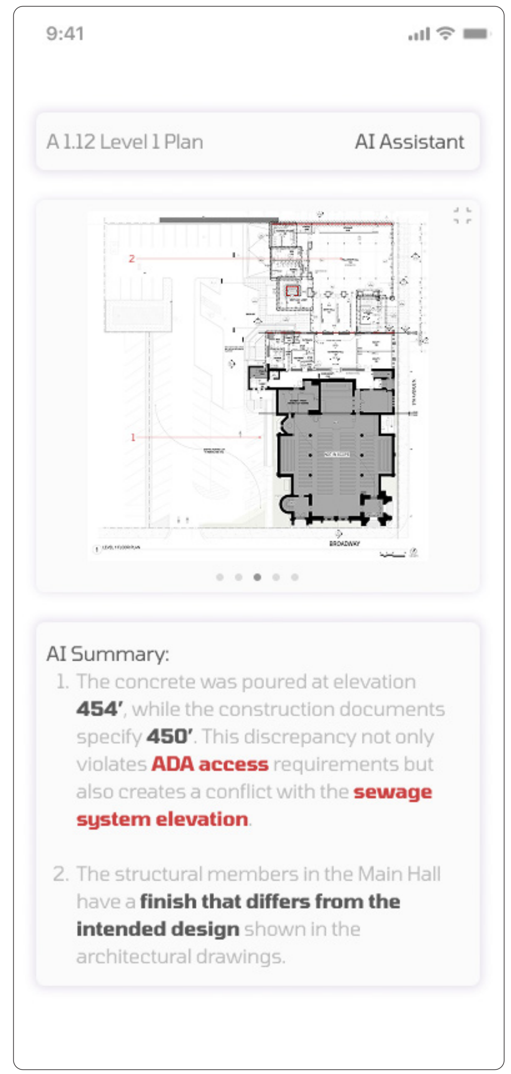


The user selects a project to begin the **construction administration** process, and the chosen project is displayed in a **clear, non-frosted view**.

What to do in the Project

The user can select the **sheet, location, and progress** of the construction project.

Past Notes

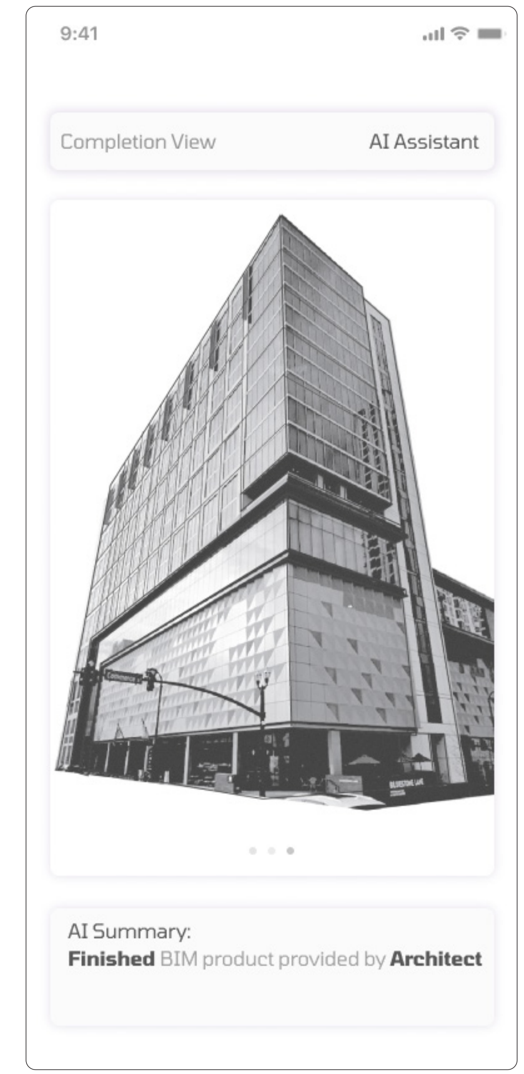


AI-generated notes are saved and **highlighted in different colors and fonts** to indicate varying **levels of importance.**

Real Time View

The real-time view **displays the current scenes on the construction site**, with a **thick border** to emphasize that it represents the physical, real-world environment.

BIM View



The **BIM view** displays the model at the current on-site location, showing the **completed elements** of the project.

Mixed Reality View

The **mixed reality view** integrates real-world construction with the **BIM model**, applying **computer vision and AI** to detect site imperfections. It automatically generates notes to **document discrepancies and mitigate future errors.**

Martin He portfolio

Thank You!

[selected Programming, HCI, and Architecture works]
Carnegie Mellon University SOA B.Arch